Empirical measures in dynamical systems

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À todos aqueles que seguem lutando por um mundo mais humano e justo.

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Chapter 1

Ergodic theory and dynamics

1.1 A categorical zoo

The starting point of the field of dynamical systems is a space X, endowed with some structure, along with some notion of "time evolution", which could be a map $X \to X$ or a family thereof, which preserves the said structure. For instance, in topological dynamics, X is usually a topological space and the time evolution is given by a continuous map, while in ergodic theory the space is endowed with the structure of a measure and the maps are measure-preserving.

A very flexible way to encompass these different notions is a *monoid object* in a category. Following [1], we can define a monoid object in a category C as a functor $M \to C$, where M is *monoid*, that is, a category with a single object. An usual monoid is just a group without the inverse operation, namely, a set M endowed with a special identity element 1_M and a multiplication $(a, b) \in X \times X \mapsto a \cdot b \in X$ satisfying

i. $1_M \cdot a = a = a \cdot 1_M$ for all $a \in M$;

ii. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in M$.

Given such a monoid M, the category composed of a single object $\bigstar$, set of morphisms $\mathrm{Hom}(\bigstar, \bigstar) = M$ and composition given by the monoid multiplication will satisfy the definition of a category.

To illustrate the definition, if $f_a, f_b \colon X \to X$ are two measure-preserving maps, then the associated monoid is the one-object category for the free monoid with two generators $\{a, b\}$, and the functor sends $\bigstar$ to the object X and words to the compositions of f_a and f_b in the category of measure-preserving maps, e.g., $abba$ is sent to $f_a \circ f_b \circ f_b \circ f_a \colon X \to X$.

This point of view also allows for situations in which the morphisms carry more structure than a single function from a set to itself satisfying some properties. A great example of this are Markov chains, whose time evolution does not map points to points but to distributions. Another example is the action of an iterated function system on closed subsets of a compact metric space, whose time evolution maps points to a closed subset — namely, the set of places where it could be if we pick one of the maps at random. Below, we will describe both with more details.

Some examples

Category of Markov kernels

Sometimes, it's useful to model uncertainty in dynamics by having the maps send each point $x \in X$ not to a point $f(x) \in X$, but to a *probability measure* $f(x) \in \mathcal{M}_1(X)$. The probability measure $f(x)$ then encodes the distribution of expected outcomes given that we started in the state x.

It may not be so clear, however, how we can compose iterates of f and regard it as a dynamic on X. Semantically, we want the iterates to encode the distribution of expected outcomes after n steps given that the system started at the point x. To work it out, we can start with the case where X is finite, so any $f(x) \in \mathcal{M}_1(X)$ is completely determined by some transition probabilities $\mathbb{P}(f(x) = y) := f(x)(\{y\})$. Then, supposing that the probability of first making the transition $x \mapsto y$ and then $y \mapsto z$ is $\mathbb{P}(f(x) = y) \cdot \mathbb{P}(f(y) = z)$, we should have

$$\mathbb{P}(f^2(x) = z) = \sum_{y \in X} \mathbb{P}(f(x) = y) \cdot \mathbb{P}(f(y) = z).$$

If we rewrite the above with a measure-theoretic language, we see that we ought to define

$$f^2(x)(E) = \int_X f(y)(E)\ d(f(x))\{y\}.$$

More generally, given $f\colon X \to \mathcal{M}_1(Y)$, $g\colon Y \to \mathcal{M}_1(Z)$, we can define a kernel composition by

$$(g \circ f)(x)(E) = \int_X g(y)(E)\ d(f(x))\{y\}$$

so that $g \circ f\colon X \to \mathcal{M}_1(Z)$.

A map with type $X \to \mathcal{M}_1(Y)$ is called a *Markov kernel* between the measurable spaces X and Y. The category whose objects are measurable spaces and morphisms are Markov kernels, with compositions given by the rule above, will be called **Stoch**. We remark that the identity morphisms in this category are the Dirac masses $\delta(x) = \delta_x \in \mathcal{M}_1(X)$.

An interesting aspect of this category is that "deterministic" measurable maps still live inside it — given $g\colon X \to Y$ a measurable function, $f_g(x) = \delta_y$ is a Markov kernel. That is to say, this correspondence defines a (faithful) functor from the category **Meas** of measurable functions to **Stoch**. This means that dynamics in **Stoch** can be seen a generalization of deterministic measurable systems, where we now also allow the maps to be stochastic.

Another observation is that a probability measure μ in a measurable space X can be represented in this category by a Markov kernel $m\colon \{\star\} \to \mathcal{M}_1(X)$, where $\{\star\}$ is a measurable space with one element. Namely, we set $m(\star) = \mu$.

Iterated function systems

Invariant states

Given categories $\mathcal{C}, \mathcal{D}$, a *cone* of a functor $F\colon \mathcal{D} \to \mathcal{C}$ is defined as an object $C \in \mathcal{C}$ and a natural transformation $T\colon \Delta(C) \Rightarrow F$, where $\Delta(C)\colon \mathcal{D} \to \mathcal{C}$ is the constant functor at the identity of C.

In the case we are interested, $\mathcal{D} = M$ is a monoid and F is given by the dynamics. This means a cone is given as an object $C \in \mathcal{C}$ and a morphism $c\colon C \to X$ with the property that for every $m \in M$,

$$\begin{tikzcd} & C \arrow[dl, "c"'] \arrow[dr, "c"] & \\ X \arrow[rr, "m"] & & X \end{tikzcd}$$

commutes.

A cone (C, c) is *universal* for $M \to \mathcal{C}$ if it is a limit of the above diagram, that is, for every other cone (D, d), there is a unique arrow $D \to C$ so that

$$\begin{tikzcd} & D \arrow[d, dashed] \arrow[ddl, "d"'] \arrow[ddr, "d"] & \\ & C \arrow[dl] \arrow[dr] & \\ X \arrow[rr, "m"] & & X \end{tikzcd}$$

commutes. Since categorical limits are unique, if a universal cone exists, it is unique up to isomorphism as well.

Example 1.1.1. Let $f\colon X \to X$ be any function from a set to itself. Like in a previous example, this will define a dynamical system in **Set** over the monoid $\mathbb{Z}_{\geq 0}$, mapping $n \in \mathbb{Z}_{\geq 0}$ to the iterate f^n. If $x \in X$ is a fixed point of f, then we get a cone by defining $C = \{x\}$ and picking the morphism $c\colon C \to X$ to be the inclusion. Since x is a fixed point, it's clear that the associated diagram will commute. Conversely, if (C, c) is a cone and $\star \in C$, then the commutativity will mean $c(\star)$ is a fixed point for f.

The *universal* cone for f, then, will correspond to the set of all fixed points (which may be empty) paired with the set inclusion. Any other cone will map uniquely to this set, by matching each of its points with the corresponding fixed point for f.

Example 1.1.2. Let $f\colon X \to \mathcal{M}_1(X)$ be a *Markov kernel* from a measurable set to itself. Examples of cones for the associated system in **Stoch** over $\mathbb{Z}_{\geq 0}$ are *stationary measures* $m\colon \{\star\} \to \mathcal{M}_1(X)$, that is, measures with the property that

$$(f \circ m)(\star) = \int_X f(x)\, d\mu\{x\} = \mu,$$

where $\mu = m(\star)$.

1.2 Extensions

references for the section [2]

Extensions are ways to see one dynamic inside another one.

The invertible extension

Solenoid

A failure of existence

Let $X = \mathbb{S}^1$ be the circle with the Borel σ-algebra $\mathcal{B}$ and $f\colon \mathbb{S}^1 \to \mathbb{S}^1$ be an irrational rotation. With the Axiom of Choice, we can form a set E with exactly one point from each orbit of f. Since f is a Lebesgue measure-preserving bijection, $E_n = f^n(E)$ are disjoint, have the same outer measure and $\bigcup_{n\in\mathbb{Z}} E_n = \mathbb{S}^1$. In particular, the E_n cannot be measurable, thus $m^*(E) > 0$.

If K is a subset of positive outer measure and $\mathcal{E} = \sigma(\{K\} \cup \mathcal{B})$, and if we define $\mu^*\colon 2^X \to [0,1]$ as

$$\mu^*(A) = \frac{m^*(A \cap K)}{m^*(K)},$$

then μ^* is an probability outer measure that restricts itself to a probability measure μ on $\mathcal{E}$. It is also the case that $\mathcal{E} = \mathcal{B} \cup \{A \cap K : A \in \mathcal{B}\} \cup \{A \cap K^c : A \in \mathcal{B}\}$. Let C_K be the completion by μ^* of the σ-algebra $\mathcal{E}$.

Let's apply the above construction for $K = \bigcup_{n=1}^{\infty} f^n(E)$. Note that $E \cap K = \varnothing$, therefore $\mu^*(E) = 0$ and $E \in C_K$. We conclude that $f^{-1}(K) = E \cup K$ is measurable as well, so for every $A \in \mathcal{B}$, $f^{-1}(A \cap K) = f^{-1}(A) \cap f^{-1}(K)$ is measurable, thus $f^{-1}(\mathcal{E}) \subseteq C_K$.

Now, for any $A \in C_K$, we can write $A = B \cup Z$ with $B \in \mathcal{E}$ and $\mu^*(Z) = 0$; thus, $f^{-1}(A) = f^{-1}(B) \cup f^{-1}(Z)$. Note that

$$\mu^*(f^{-1}(Z)) = m^*(Z \cap f(K)) \le m^*(Z \cap K) = \mu^*(Z) = 0,$$

therefore $f^{-1}(C_K) \subset C_K$.

The same calculation shows that $\mu^*(f^{-1}(A)) \le \mu^*(A)$ for any $A \subseteq X$. Thus, if A is measurable,

$$1 - \mu(f^{-1}(A)) = \mu(f^{-1}(A^c)) \le \mu(A^c) = 1 - \mu(A),$$

so $f_*\mu = \mu$ on C_K.

Consider $\Sigma = \{(x_i)_{i\le 0} : f(x_i) = x_{i+1}\} \subseteq X^{\mathbb{Z}_{\le 0}}$ as the inverse limit of (X, f, C_K), endowed with the restriction of the product σ-algebra. Note that $\tilde{f}\colon \Sigma \to \Sigma$ given by

$$\tilde{f}((x_i)_{i\le 0}) = (f(x_i))_{i\le 0} = (\ldots, x_{-1}, x_0, f(x_0))$$

is invertible and measurable, since

$$\tilde{f}([A_{-k}, \ldots, A_0] \cap \Sigma) = [A_{-k}, \ldots, A_0, X] \cap \Sigma.$$

For each $n \in \mathbb{Z}_{\le 0}$, let $K_n = [n \mid K] = \{(x_i)_{i\le 0} : x_n \in K\}$. Suppose there exists $\tilde{\mu}$, a measure on Σ such that $\pi_*\tilde{\mu} = \mu$, where $\pi((x_i)_{i\le 0}) = x_0$. Then

$$\tilde{\mu}(K_n) = \tilde{\mu}(\tilde{f}^n(\pi^{-1}(K))) = \mu(K) = 1$$

for every n. On the other hand, since f is invertible,

$$\begin{aligned}\bigcap_{n\le 0} K_n &= \{(x_i)_{i\le 0} : \forall i, x_i \in K\} \\ &= \{(x_i)_{i\le 0} : \forall n \ge 0, f^{-n}(x_0) \in K\}.\end{aligned}$$

But this set is empty because

$$\{(x_i)_{i\leq 0} : \forall n \geq 0, f^{-n}(x_0) \in K\} = \bigcap_{n=0}^{\infty} f^n(K) = \limsup_n E_n = \varnothing.$$

1.3 Cocycles and skew products

Example 1.3.1 (Iterated function system)**.**

Lyapunov exponents

1.4 Empirical measures

Given a map $f\colon X \to X$ on a measurable space and a point $x \in X$, we can define a sequence of *empirical measures*

$$e_n(x) = \frac{1}{n}\sum_{i=0}^{n-1} \delta_{f^i(x)},$$

encoding the averages along the orbit up to a time n. These measures are, indeed, uniquely defined by the property that for any observable $\phi\colon X \to \mathbb{R}$, its integral with respect to $e_n(x)$ equals the Birkhoff average up to time n:

$$\int \phi \; d(e_n(x)) = \frac{1}{n}\sum_{i=0}^{n-1} \phi(f^i(x)),$$

and it is this quality of being defined by the values observed on a finite number of steps along an orbit that should justify the nomenclature empirical. The direct relation to Birkhoff averages also highlights these objects as having a central role in ergodic theory.

It's not always the case that this sequence converges (weakly) to some measure, but when it does, the limit is invariant and represents the asymptotic distribution of the orbit. Reciprocally, For each invariant measure $\mu \in \mathcal{M}_f(X)$, we can ask whether the empirical measures of a point will converge weakly to it. The set $\mathcal{B}(\mu) \subseteq X$ of points with this property is called the *basin* of the measure μ.

It may happen that the set of points whose empirical measures do not converge is large, in the sense that it has positive volume. Systems with this property are said to exhibit *historical behavior*, and the existence of diffeomorphisms exhibiting historical behavior in a robust way is an important problem in dynamics.

Proposition 1. *Let X be a compact metric space with the Borel σ-algebra. For every $x \in X$, the limit points in the weak* topology of the sequence of empirical measures*

$$\mathcal{M}(x) = \{e_n(x)\}'_{n\geq 0} = \bigcap_{i=0}^{\infty} \overline{(\{e_n(x)\}_{n\geq i})}$$

form a compact connected subset of the invariant measures.

Proof. The set is closed by definition, so by Banach-Alaoglu it's compact; for connectedness, suppose $\mathcal{M}(x) = K_1 \cup K_2$ where K_1, K_2 are closed and disjoint. Let $\{\phi_i\}_{i\in\mathbb{N}} \subseteq B_{C(X)}$ be a dense subset and define

$$d(\mu, \nu) = \sum_{i=1}^{\infty} \frac{|\mu(\phi_i) - \nu(\phi_i)|}{2^i},$$

which is a metric inducing the weak* topology on the set of probability measures. Since it's a metric, there are two open sets $U_1 \supseteq K_1, U_2 \supseteq K_2$ with positive distance $s > 0$ from each other. We claim $\lim_{n\to\infty} d(e_n(x), e_{n+1}(x)) = 0$, which would contradict $s > 0$ because the definition of $\mathcal{M}(x)$ implies $e_n(x) \in U_1$ and $e_{n+1}(x) \in U_2$ for infinitely many n. Indeed,

$$\begin{aligned} d(e_n(x), e_{n+1}(x)) &= \sum_{i=1}^{\infty} \frac{|\frac{1}{n}\sum_{i=0}^{n-1} \phi_i \circ f^i(x) - \frac{1}{n+1}\sum_{i=0}^{n} \phi_i \circ f^i(x)|}{2^i} \\ &\leq \sum_{i=1}^{\infty} \frac{2}{n}\frac{\|\phi_i\|_\infty}{2^i} \leq \frac{2}{n} \to 0. \end{aligned}$$

□

1.5 Ergodic theorems

Below we reproduce a succinct proof of the pointwise ergodic theorem, described for example in [3, p. 136][1].

Theorem 2 (Birkhoff Ergodic Theorem)**.** *Let μ be an invariant probability measure and $\phi\colon X \to \mathbb{R}$ an integrable function. Then, for μ-almost every $x \in X$,*

$$\frac{1}{n}\sum_{i=0}^{n-1} \phi(f^i(x)) \to \mathbb{E}_\mu(\phi \mid \mathcal{I})(x),$$

where $\mathcal{I}$ is the sub-σ-algebra of (strictly) invariant sets.

Proof. Let

$$F_n(x) = \max_{1\leq k\leq n} \sum_{i=0}^{k} \phi(f^i(x)).$$

Claim1: $F_{n+1} = \phi + \max\{0, F_n \circ f\}$.

To see this, note that $\sum_{i=0}^{k} \phi \circ f^i = \phi + \sum_{i=0}^{k-1} \phi \circ f^{i+1}$, so

$$F_{n+1} = \phi + \max_{1\leq k\leq n+1} \sum_{i=0}^{k-1} \phi \circ f^{i+1} = \phi + \max\left\{0, \max_{1\leq k\leq n} \sum_{i=0}^{k} \phi \circ f^{i+1}\right\} = \phi + \max\{0, F_n \circ f\}.$$

Claim2: $F_{n+1} - F_n \circ f = \phi - \min\{0, F_n \circ f\}$.

[1] This is not exactly relevant to this dissertation, but the author has formally verified this proof in the Lean 4 ITP, and the code is currently available at `https://github.com/soficshift/pointwise-birkhoff`.

This claim follows from the previous one by rewriting

$$F_{n+1} + \min\{0, F_n \circ f\} = \phi + \max\{0, F_n \circ f\} + \min\{0, F_n \circ f\} = \phi + F_n \circ f.$$

Two very useful corollaries of the claim are that for each $x \in X$, $F_{n+1}(x) - F_n(f(x))$ is a non-increasing sequence (because $F_n(f(x))$ is non-decreasing by definition), and that if $F_n(f(x)) \to \infty$, then the sequence will converge monotonically to $\phi(x)$.

Now, we define the set

$$S = \{x \in X \mid F_n(x) \to +\infty\}.$$

The first observation is that $S \in \mathcal{I}$, because it's measurable and strictly invariant. Since μ is invariant by f, this implies $\int_S F_n \circ f \, d\mu = \int_S F_n \, d\mu$. Also, the observations from the last claim show that $F_{n+1}(x) - F_n(f(x)) \to \phi(x)$ for $x \in S$. Thus, by the dominated convergence theorem,

$$0 \leq \int_S F_{n+1} - F_n \, d\mu = \int_S F_{n+1} - F_n \circ f \, d\mu \to \int_S \phi \, d\mu \tag{1.1}$$

which means $\int_S \phi \, d\mu \geq 0$.

The second observation is that if $x \notin S$, then $(F_n(x))_n$ is bounded above, so

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \leq \limsup_{n\to\infty} \frac{1}{n} F_n(x) \leq 0. \tag{1.2}$$

Putting the two together, we conclude that if ϕ is such that $\mathbb{E}(\phi \mid \mathcal{I}) < 0$ almost everywhere, then by eq. (1.1) and the fact $S \in \mathcal{I}$, the set S has measure zero, so eq. (1.2) is valid almost everywhere.

The theorem then follows by noticing that for any $\varepsilon > 0$, if we replace ϕ by $\tilde{\phi} = \phi - \mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon$, then $\mathbb{E}(\tilde{\phi} \mid \mathcal{I}) < 0$, so

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \tilde{\phi}(f^i(x)) = \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) - \mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon \leq 0.$$

If we combine this with the analogous inequality for $-\phi$, we get

$$\mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon \leq \liminf_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \leq \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \leq \mathbb{E}(\phi \mid \mathcal{I}) + \varepsilon$$

and the result follows by letting $\varepsilon \to 0$. □

1.6 Partitions

A *partition* of a set S is a collection of disjoint subsets $\mathcal{P} \subseteq 2^S$ whose union is S. It's important to notice that $\mathcal{P}$ can be seen either as a set of subsets or (with some abuse of notation) as a function $\mathcal{P} \colon S \to 2^S$ assigning to each point $p \in S$ the unique $\mathcal{P}(p) \in \mathcal{P}$ such that $p \in \mathcal{P}(p)$. To make matters clearer, henceforward we shall denote by $\overline{\mathcal{P}} \subseteq 2^S$ the image of $\mathcal{P} \colon S \to 2^S$ whenever we need to emphasize that the set of subsets is the object of our attentions.

Lattice structure

The partitions $\mathbf{P}(S)$ of a set S form a lattice under the refinement order. Namely, we endow $\mathbf{P}(S)$ with the order where $\mathcal{P} \preceq \mathcal{Q}$ if and only if for all $Q \in \overline{\mathcal{Q}}$, there is some $P \in \overline{\mathcal{P}}$ such that $Q \subseteq P$, and whenever this happens we say that $\mathcal{Q}$ is *finer* and $\mathcal{P}$ is *coarser* relative to each other. Given two partitions $\mathcal{P}, \mathcal{Q} \in \mathbf{P}(S)$, there is a *join* operation defined by

$$(\mathcal{P} \vee \mathcal{Q})(x) = \mathcal{P}(x) \cap \mathcal{Q}(x)$$

that is the least upper bound of $\mathcal{P}$ and $\mathcal{Q}$ (the coarsest partition finer than both $\mathcal{P}$ and $\mathcal{Q}$). There is also a *meet* operation $\mathcal{P} \wedge \mathcal{Q}$, giving the greatest lower bound, but it is less straightforward to construct: on the union $\overline{\mathcal{P}} \cup \overline{\mathcal{Q}}$, we say P and Q are related if they overlap, that is, $P \cap Q \neq \varnothing$, and taking the transitive closure we get an equivalence relation. The elements of $\mathcal{P} \wedge \mathcal{Q}$, then, are precisely the unions of sets in the same equivalence class under this relation. The meet operation will not be used in the following text, though.

The partitions $\mathbf{P}(S)$ form, in fact, a *complete* lattice, meaning that all subsets $P \subseteq \mathbf{P}(S)$ have both a least upper bound $\bigvee_{\mathcal{P} \in P} \mathcal{P} \in \mathbf{P}(S)$ and a greatest lower bound $\bigwedge_{\mathcal{P} \in P} \mathcal{P} \in \mathbf{P}(S)$. Their construction is very similar to their binary versions: the join can be defined as $\left(\bigvee_{\mathcal{P} \in P} \mathcal{P}\right)(x) = \bigcap_{\mathcal{P} \in P} \mathcal{P}(x)$, and we can use the join to define the meet as

$$\bigwedge_{\mathcal{P} \in P} \mathcal{P} = \bigvee_{\substack{\mathcal{Q} \text{ is an upper} \\ \text{bound for } P}} \mathcal{Q}.$$

A measured interlude

In a measure space, it is often useful to relax the notions of equality and order on the partitions so that zero measure sets are disregarded. This is done in the obvious way, namely by saying $\mathcal{P} \preceq \mathcal{Q}$ if for all $Q \in \overline{\mathcal{Q}}$, there is a $P \in \overline{\mathcal{P}}$ such that $Q \setminus P$ has zero measure. Going a bit further, the partitions themselves may be allowed to cover only a total measure subset of the space, and their elements could overlap on zero measure sets. Again — always in the most natural way, and we shall henceforward hide the petty details in the hopes the reader is able to extrapolate them.

1.7 Disintegrating measures

In the Calculus course, we learn that it's most often easier to find the area of a region in the plane if we reduce the problem back to a single dimension, by taking integrals of the length along parallel lines. This is the essence of Fubini's theorem, which states that the integral of a function in a higher dimensional region can be written as nested integrals of the function along one-dimensional subsets. Then we go to the probability course, and the lecturer talks about conditional expectations and the whole philosophy of Bayes of taking probabilities conditioned to some event. It may not seem at first that this has anything to do with the aforementioned theorem in the Calculus course, at least not until we take the Ergodic theory course and we learn about measures and disintegrations.

While Fubini remains true for product measures, in general the problem of decomposing an arbitrary measure into a family of measures defined in smaller subsets is not as straightforward. Neither is straightforward to see how could one talk about the probability of an event conditioned to another event of *zero probability*, even though in practice this happens very often. Both situations are instances of the general problem the disintegration of measures solves: given a measure μ and a partition $\mathcal{P}$ of the space X, we want to find a family of measures $\{\mu_x\}_{x\in X}$, each μ_x giving full measure to $\mathcal{P}(x)$, with the property that $\mu(E) = \int \mu_x(E)\, d\mu(x)$.

Definition 1 (Disintegration)**.** A *disintegration* of a measure $\mu \in \mathcal{M}(X)$ with respect to a partition $\mathcal{P}$ is a family $\{\mu_x\}_{x\in X} \subseteq \mathcal{M}(X)$ such that for every measurable set $E \subseteq X$,

1. $\mu_x(\mathcal{P}(x)) = 1$ for μ-almost every x;
2. the map $x \mapsto \mu_x(E)$ is measurable;
3. $\mu(E) = \int \mu_x(E)\, d\mu(x)$.

Uniqueness

Measurability

Not all partitions are created equal. Some may be pathological enough that there may be no hope in disintegrating over them. In fact, such partitions appear very *naturally* in dynamics, namely as the orbits of points of many maps with an invariant measure.

Example 1.7.1. Suppose $f\colon X \to X$ is invertible and μ is an invariant measure for f such that $\mu(X \setminus \mathrm{Per}(f)) > 0$. Then, the partition $\mathcal{O}_f\colon X \to 2^X$ of two-sided orbits of f is such that μ does not admit a disintegration with respect to $\mathcal{O}_f$.

Proof. Suppose that such disintegration $\{\mu_x\}_{x\in X}$ exists. We claim that $\{f_*\mu_x\}_{x\in X}$ is also a disintegration. Indeed, all $\mathcal{O}_f(x)$ are invariant, so we have

$$(f_*\mu_x)(\mathcal{O}_f(x)) = \mu_x(f^{-1}(\mathcal{O}_f(x))) = \mu_x(\mathcal{O}_f(x)) = 1$$

for a.e. x, and similarly, we have that $x \mapsto (f_*\mu_x)(E) = \mu_x(f^{-1}(E))$ is measurable, and $\mu(E) = \mu(f^{-1}(E)) = \int f_*\mu_x(E)\, d\mu(x)$. By uniqueness, this implies that $f_*\mu_x = \mu_x$ almost everywhere.

By item 1, the hypothesis and the paragraph above, we can find some $x \notin \mathrm{Per}(f)$ such that $f_*\mu_x = \mu_x$ and $\mu_x(\mathcal{O}_f(x)) = 1$. But this is impossible: since $\mathcal{O}_f(x)$ is infinite and $\mu_x(\{f^k(x)\}) = \mu_x(\{x\})$ for every k,

$$1 = \mu_x(\mathcal{O}_f(x)) = \sum_{k\in\mathbb{Z}} \mu_x(\{f^k(x)\}) = \sum_{k\in\mathbb{Z}} \mu_x(\{x\}) \in \{0, \infty\}.$$

□

Remark 1. A very basic instance of example 1.7.1 would be with $X = S^1$, $f\colon S^1 \to S^1$ an irrational rotation, and μ the unique ergodic measure, because f has no periodic points.

Remark 2. The example 1.7.1 also shows that it is not sufficient that the *elements* of the partition are measurable sets. In the remark above, the orbits are countable sets of points in the circle, and thus are Lebesgue (or even Borel) measurable, and yet the partition does not admit a disintegration.

The right notion of a "well-behaved" partition with respect to disintegrations will turn out to be the notion of *measurable partitions*.

Definition 2. A partition $\mathcal{P}$ is said to be *measurable* if there is a sequence $\mathcal{P}_1, \mathcal{P}_2, \ldots$ of partitions, all of which composed of a countable number of measurable sets, such that $\mathcal{P} = \bigvee_{n\in\mathbb{N}} \mathcal{P}_n$.

Example 1.7.2. The partition $\perp$ of all points of a compact metric space X is measurable in the Borel σ-algebra.

Proof. Let $\{U_n\}_{n\in\mathbb{N}}$ be a countable basis for the topology of X, and define $\mathcal{P}_n = \{U_n, U_n^c\}$. Let $P \in \mathcal{P} = \bigvee_{n\in\mathbb{N}} \mathcal{P}_n$, which is measurable by definition, and $p \in P$. If $q \in P$, then we must have $\mathcal{P}_n(p) = \mathcal{P}_n(q)$ for all $n \in \mathbb{N}$, or in other words, $p \in U_n \iff q \in U_n$, for all n. The T_1 property then implies $p = q$, so $P = \{p\}$. This shows $\mathcal{P} = \perp$. □

Example 1.7.3. Any partition $\mathcal{P}$ of a compact metric space X whose elements are closed is measurable in the Borel σ-algebra.

Proof. Since they are closed, the elements of $\mathcal{P}$ are measurable in the Borel algebra. To show the partition is countably generated, we can define a metric on $\overline{\mathcal{P}}$ by letting $d(P_1, P_2)$ be the distance between the compact sets $P_1, P_2 \in \overline{\mathcal{P}}$ (this is the Hausdorff metric, which we detail in **??**). With this metric, the surjection $\mathcal{P} : X \to \overline{\mathcal{P}}$ is a contraction, so $\overline{\mathcal{P}}$ is a compact metric space as well. By the example 1.7.2, the partition $\perp$ of points of $\overline{\mathcal{P}}$ is countably generated, thus so is its preimage by $\mathcal{P}$. □

Construction of conditional measures

1.8 Entropy

Topological entropy

Measure-theoretic entropy

Pesin entropy formula

Chapter 2

Schwarzian derivatives and cylinder maps

In this chapter, we follow the exposition by Bonifant and Milnor [4] on a family of endomorphisms of a cylinder.

2.1 Cross ratios and the Schwarzian derivative

Suppose we start with a field $\mathbb{K}$ and, not being happy with the fact that we cannot divide by zero, decide to add a point at infinity; this gives us the *projective line* $P^1(\mathbb{K}) = \mathbb{K} \sqcup \{\infty\}$, whose arithmetic operations are the same as those of $\mathbb{K}$ except for the additional rules $\frac{1}{0} = \infty$, $\frac{1}{\infty} = 0$, $x \cdot \infty = \infty$ if $x \neq 0$, and $x + \infty = \infty$ if $x \neq \infty$. It's not possible to define such operations everywhere without introducing inconsistent rules, so the operations are partial — we cannot multiply $\infty \cdot 0$ or add $\infty + \infty$.

Our new space $P^1(\mathbb{K})$ has an interesting property: given $z \in P^1(\mathbb{K})$, we have three types of operations that we can apply to z: translations $z \mapsto z + x$, where $x \in \mathbb{K}$, homotheties $z \mapsto xz$ for $x \in \mathbb{K} \setminus \{0\}$, and the inverse $z \mapsto 1/z$, which was not defined before if $z = 0$, but now makes sense for every z. These three families of actions, which are invertible, generate the *group of homographies* with coefficients in $\mathbb{K}$.

A second way to construct $P^1(\mathbb{K})$, which makes the role of the group of homographies as a symmetry group of this space even clearer, is to think of the space of "formal fractions", that is, pairs of numbers (x, y), where $x, y \in \mathbb{K}$ are not both zero, where we identify "equivalent fractions", in the sense that $(\lambda x, \lambda y) \sim (x, y)$ for every $\lambda \in \mathbb{K} \setminus \{0\}$. The quotient by this equivalence relation gives us $P^1(\mathbb{K})$ as the space of lines passing through the origin of the vector space $\mathbb{K}^2$, and the operations are like fraction operations: $(x, y) + (z, w) = (wx + yz, yw)$, $(x, y) \cdot (z, w) = (xz, yw)$, and $(x, y)^{-1} = (y, x)$. Despite being seemingly more complicated, the interesting fact about this construction is that it emphasizes how the symmetry group of the space $\mathbb{K}^2$, the linear group $\mathrm{GL}_2(\mathbb{K})$, naturally induces a symmetry group on the quotient, the projective group $\mathrm{PGL}_2(\mathbb{K})$, which is precisely the group of homographies.

This group is a quotient of $\mathrm{GL}_2(\mathbb{K})$ by a subgroup of dimension 1, so it has dimension 3. Furthermore, it is 3-transitive: given two sets of three distinct points in $P^1(\mathbb{K})$, there exists an element $g \in \mathrm{PGL}_2(\mathbb{K})$ that takes the first set to the second. But not only that, $P^1(\mathbb{K})$ has three special points: 0, 1, and ∞. Thus, given $x_1, x_2, x_3 \in P^1(\mathbb{K})$ distinct, we can ask which $g \in \mathrm{PGL}_2(\mathbb{K})$ such that $g(x_1) = \infty$, $g(x_2) = 0$, and $g(x_3) = 1$. With some calculations, we arrive at

$$g(z) = \frac{(x_2 - z)(x_3 - x_1)}{(x_1 - z)(x_3 - x_2)}$$

The *cross ratio* of four points x_0, x_1, x_2, x_3 is the value of $g(x_0)$ for the element g of the homography group that sends $x_1 \mapsto \infty$, $x_2 \mapsto 0$, and $x_3 \mapsto 1$.

Definition 3 (Cross ratio)**.**

$$\rho(x_0, x_1, x_2, x_3) = \frac{(x_2 - x_0)(x_3 - x_1)}{(x_1 - x_0)(x_3 - x_2)}$$

Definition 4. Let $f : I \to I'$ be a C^3 diffeomorphism between two intervals of the real line. The *Schwarzian derivative* of f is the function

$$\mathcal{S}f = \frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2.$$

We shall write $\mathcal{S}f \prec 0$ if there exists a dense open set $U \subseteq I$ such that $\mathcal{S}f(y) < 0$ for every $y \in U$. Similarly, we write $\mathcal{S}f \succ 0$ if there exists $U \subseteq I$ open and dense such that $\mathcal{S}f(y) > 0$ for every $y \in U$.

The Schwarzian derivative has some important properties that we highlight in the Proposition below, vide [4, Proposition B.1].

Proposition 3. *The Schwarzian derivative has the following properties:*

1. *If $\mathcal{S}f \succ 0$ and $\mathcal{S}g \succ 0$, then $\mathcal{S}(f \circ g) \succ 0$, and similarly for $\prec 0$. Moreover, if $\mathcal{S}f \equiv 0$, then $\mathcal{S}(f \circ g) = \mathcal{S}(g)$.*
2. *If $\mathcal{S}f \prec 0$, then $\mathcal{S}f^{-1} \succ 0$.*
3. *$\mathcal{S}f \prec 0$ if, and only if, $\varphi(y) = 1/\sqrt{|f'(y)|}$ is strictly convex.*
4. *$\mathcal{S}f \equiv 0$ if, and only if, f is a fractional linear transformation.*

Proof.

1. We have

$$\mathcal{S}(f \circ g) = \frac{(f \circ g)'''}{(f \circ g)'} - \frac{3}{2}\left(\frac{(f \circ g)''}{(f \circ g)'}\right)^2.$$

By the Faà di Bruno's formula [5, p. 479] in its combinatorial form,

$$(f \circ g)^{(n)}(x) = \sum_{\pi \in \Pi} f^{(|\pi|)}(g(x)) \cdot \prod_{B \in \pi} g^{(|B|)}(x)$$

where the sum is taken over the set Π of all partitions of $\{1, \ldots, n\}$, and $B \in \pi$ means B is an element of the partition π (therefore, a subset of $\{1, \ldots, n\}$).

Specializing Faà di Bruno's formula to $n = 1, 2, 3$, we have

$$\begin{aligned}(f \circ g)' &= (f' \circ g) \cdot g' \\ (f \circ g)'' &= (f'' \circ g) \cdot (g')^2 + (f' \circ g) \cdot g'' \\ (f \circ g)''' &= (f''' \circ g) \cdot (g')^3 + 3(f'' \circ g) \cdot g'' \cdot g' + (f' \circ g) \cdot g'''\end{aligned}$$

and thus

$$\begin{aligned}\mathcal{S}(f \circ g) &= \frac{(f''' \circ g) \cdot (g')^3 + 3(f'' \circ g) \cdot g'' \cdot g' + (f' \circ g) \cdot g'''}{(f' \circ g) \cdot g'} \\ &\qquad - \frac{3}{2}\left(\frac{(f'' \circ g) \cdot (g')^2 + (f' \circ g) \cdot g''}{(f' \circ g) \cdot g'}\right)^2 \\ &= \frac{(f''' \circ g) \cdot (g')^2}{f' \circ g} + 3\frac{(f'' \circ g) \cdot g''}{f' \circ g} + \frac{g'''}{g'} - \frac{3}{2}\left(\frac{(f'' \circ g) \cdot g'}{f' \circ g} + \frac{g''}{g'}\right)^2 \\ &= \left(\frac{f''' \circ g}{f' \circ g} - \frac{3}{2}\left(\frac{f'' \circ g}{f' \circ g}\right)^2\right) \cdot (g')^2 + \frac{g'''}{g'} - \frac{3}{2}\left(\frac{g''}{g'}\right)^2 \\ &= (\mathcal{S}f) \circ g \cdot (g')^2 + \mathcal{S}g. \qquad (2.1)\end{aligned}$$

Since g is a diffeomorphism, $g'(x) \neq 0$ for all $x \in I$, and therefore $(g')^2 > 0$. In particular, g is a homeomorphism, so that if $\mathcal{S}f(x) > 0$ for all x the open dense set U, and $\mathcal{S}g(x) > 0$ in the open dense set V, then $\mathcal{S}(f \circ g) > 0$ in the open dense set $g^{-1}(U) \cap V$. Similarly, from eq. (2.1) we see that if $\mathcal{S}f \equiv 0$, then the first term is zero, therefore $\mathcal{S}(f \circ g) = \mathcal{S}g$.

2. Applying eq. (2.1) to $\mathrm{id} = f^{-1} \circ f$, we see that

$$0 = \mathcal{S}(\mathrm{id}) = (\mathcal{S}f^{-1}) \circ f \cdot (f')^2 + \mathcal{S}f,$$

and by the same argument of the paragraph above, we conclude $\mathcal{S}f \prec 0 \iff \mathcal{S}f^{-1} \succ 0$.

3. Since f is a diffeomorphism between intervals, either $f' > 0$ or $f' < 0$. As a first step, assume $f' > 0$. Taking the second derivative of $\varphi(y) = 1/\sqrt{f'(y)} = f'(y)^{-1/2}$,

$$\varphi'' = \left(-\frac{1}{2}(f')^{-3/2} \cdot f''\right)' = \frac{3}{4}(f')^{-5/2} \cdot (f'')^2 - \frac{1}{2}(f')^{-3/2} \cdot f'''.$$

Thus

$$2\varphi''/\varphi = \frac{3}{2}(f')^{-4/2} \cdot (f'')^2 - (f')^{-2/2} \cdot f''' = \frac{3}{2}\left(\frac{f''}{f'}\right)^2 - \frac{f'''}{f'} = -\mathcal{S}f. \qquad (2.2)$$

From this, and noticing that $\varphi > 0$, we see that if $\mathcal{S}f(x) < 0$ if and only if $\varphi''(x) > 0$. A C^2 function $\varphi : I \to \mathbb{R}$ will be strictly convex if and only if $\varphi'' \geq 0$ and φ'' does not vanish on any open interval, so $\mathcal{S}f \prec 0$ implies that ϕ'' is positive in an open dense set, thus strictly convex. Reciprocally, if φ is strictly convex, then the set $\{x \mid \varphi''(x) > 0\} = \{x \mid \mathcal{S}f(x) < 0\}$ is open and its complement does not contain any open interval, and therefore is dense.
If $f' < 0$, we define $g\colon I \to -I'$, $g(y) = -f(y)$, so that $g' > 0$ and we fall back to the previous case. The argument is finished by noticing that $\mathcal{S}g = \mathcal{S}f$ and $1/\sqrt{|f'|} = 1/\sqrt{|g'|}$.

4. From eq. (2.2), we see that $\mathcal{S}f \equiv 0 \iff \varphi'' \equiv 0$. This, in turn, is equivalent to φ being linear, say $\varphi(y) = cy + d$. We thus have
$$f'(y) = \frac{1}{(cy+d)^2}.$$
Integrating both sides, by the Fundamental Theorem of Calculus we have
$$f(y) - f(a) = \int_a^y \frac{1}{(ct+d)^2}\,dt = \frac{1}{c}\left(\frac{1}{ca+d} - \frac{1}{cy+d}\right) = \frac{y-a}{c(ca+d)y + d(ca+d)}$$
which is a fractional linear transformation, as desired.

□

Lemma 4. *If $f\colon I \to I'$ is C^3 and $\mathcal{S}f \prec 0$, then f has at point three fixed points. When it has three fixed points, the middle one is strictly repelling and the other two are strictly attracting.*

Lemma 5. *If $f\colon I \to I'$ is C^3, then $\mathcal{S}f \prec 0$ if, and only if, f increases cross ratios, in the sense that $\rho(f(y_1), f(y_2), f(y_3), f(y_4)) > \rho(y_1, y_2, y_3, y_4)$ for all $y_1 < y_2 < y_3 < y_4$.*

Proof. Given $y_1 < y_2 < y_3 < y_4$, let φ be the linear fractional transformation that sends $f(y_1)$ to y_1, $f(y_2)$ to y_2 and $f(y_4)$ to y_4. Since from proposition 3, we have that $\mathcal{S}(\varphi \circ f) = \mathcal{S}f$, and we know fractional linear transformations also preserve cross-ratios, without loss of generality we may assume that f itself fixes y_1, y_2 and y_4. Now, by lemma 4, $f(y) > y$ on the interval (y_2, y_4), and therefore $f(y_3) > y_3$. It thus follows easily by the definition of cross-ratios that $\rho(f(y_1), f(y_2), f(y_3), f(y_4)) > \rho(y_1, y_2, y_3, y_4)$. □

Lemma 6. *If $f\colon I \to I$ preserves orientation and $\mathcal{S}f \prec 0$, then $f'(0)f'(1) < 1$.*

Proof. □

Note that by proposition 3, analogous results to the last two lemmas hold if $\mathcal{S}f \succ 0$. For example, if $\mathcal{S}f \succ 0$ then f decreases cross ratios and $f'(0)f'(1) > 1$.

2.2 Skew products

We call a map $F\colon X \times Y \to X \times Y$ a *skew product* if it is of the form $F(x, y) = (g(x), f_x(y))$ for some $g\colon X \to X$ and a family of functions $\{f_x\}_{x\in X}$, $f_x\colon Y \to Y$. We are interested in the case where X is a compact metric space with the Borel σ-algebra, $g\colon X \to X$ is a continuous map preserving an ergodic probability measure $\mu \in \mathcal{M}(X)$, $Y = I$ is the interval $[0, 1]$, and $f_x\colon I \to I$ are C^3 diffeomorphisms preserving orientation. We will see how, depending on the Schwarzian derivative of the maps f_x, we observe different behaviors of the dynamics with respect to the invariant measures $\nu_0 = \mu \otimes \delta_0$ and $\nu_1 = \mu \otimes \delta_1$. Note that for $i \in \{0, 1\}$, $\operatorname{supp} \nu_i = \mathcal{A}_i$, where $\mathcal{A}_i = X \times \{i\}$ are the "caps" of $X \times I$. Let $\mathcal{B}_i = \mathcal{B}(\nu_i)$ be the basins of the respective measures.

To simplify notation, let's denote by $f_x^n(y)$ the second coordinate of $F^n(x, y)$, that is, $f_x^n = f_{g^{n-1}(x)} \circ \cdots \circ f_x$.

Lemma 7. *If $(x, y_1) \in \mathcal{B}_0$, then $(x, y_2) \in \mathcal{B}_0$ for all $y_2 \leq y_1$. Similarly, if $(x, y_1) \in \mathcal{B}_1$, then $(x, y_2) \in \mathcal{B}_0$ for all $y_2 \geq y_1$.*

Proof. We will prove the case $i = 0$, as the situation is symmetric. Let $(x, y_1) \in \mathcal{B}_0$ and $y_2 \leq y_1$. If $\varphi \in C(X \times I)$, and $\varepsilon > 0$, by the uniform continuity of φ, there exists $\delta > 0$ such that $y < \delta$ implies $|\varphi(x, 0) - \varphi(x, y)| < \varepsilon$ for all $x \in X$. Since $U = X \times [0, \delta)$ is a continuity set for ν_0 and $e_n(x, y_2) \to \nu_0$, by Portmanteau

$$\lim_{n\to\infty} e_n(x, y_2)(U) = \nu_0(U) = 1.$$

Define $\tau_n = e_n(x, y_2)(U) = \frac{1}{n}\#\{0 \leq i < n \mid f_x^n(y_2) < \delta\}$. Note that if $f_x^n(y_2) < \delta$, then $f_x^n(y_1) < \delta$ since the diffeomorphisms preserve orientation. Thus,

$$\begin{aligned} |e_n(x, y_1)(\varphi) - e_n(x, y_2)(\varphi)| &\leq \frac{1}{n}\sum_{i=0}^{n-1}\left|\varphi(g^n(x), f_x^n(y_1)) - \varphi(g^n(x), f_x^n(y_2))\right| \\ &\leq \varepsilon\tau_n + 2\|\varphi\|_\infty(1 - \tau_n). \end{aligned}$$

Since $\tau_n \to 1$, there exists N such that $n \geq N$ implies that the right side is less than 2ε, hence both integrals on the left converge to $\nu_0(\varphi)$. □

2.3 Transversal Lyapunov exponents

To each point (x, y) of either boundary $\mathcal{A}_i$, for $i \in \{0, 1\}$, we can associate a number that expresses the average expansion rate of tangent vectors along the direction normal to the skew product boundary. This is a specific case of the general definition of Lyapunov exponents, whose existence is given by the Oseledets theorem.

Definition 5 (Transversal Lyapunov exponent)**.**

$$\mathrm{Lyap}_{\mathcal{A}_i}(x) = \lim_{n\to\infty} \frac{1}{n}\log\left|\partial_y F^n(x, i)\right|.$$

By the product rule, and noting that $\{0, 1\}$ are fixed points of the fiber diffeomorphisms, the right side of the equation above can be rewritten as

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1}\log\left|\partial_y F(g^k(x), i)\right| = \lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1}\log f'_{g^k(x)}(i).$$

Thus, by the ergodicity of μ for g and the Birkhoff ergodic theorem, for μ-almost every $x \in X$ it holds that

$$\mathrm{Lyap}_{\mathcal{A}_i}(x) = \int \log f'_x(i)\, d\mu(x), \tag{2.3}$$

and we shall denote this common value by $\mathrm{Lyap}(\mathcal{A}_i)$.

The version of the statement below is slightly more general than the corresponding one in [4, lema A.1], that asks for F to be C^2. In our context, $F(x, y) = (g(x), f_x(y))$ where g is an ergodic map for μ and $f_x \in C^1[0, 1]$ for all x. In particular, if $(x, y) \mapsto f'_x(y)$ is continuous, the following equicontinuity hypothesis is satisfied.

Lemma 8. *If the family $\{f_x'\}_{x\in X}$ is equicontinuous at $i \in \{0,1\}$ and* $\mathrm{Lyap}(\mathcal{A}_i) < 0$*, then the basin of $\mu \otimes \delta_i$ has positive $\mu \otimes \mathrm{Leb}$ measure. Conversely, if* $\mathrm{Lyap}(\mathcal{A}_i) > 0$*, then the corresponding basin has null measure.*

Proof. We will assume $i = 0$. Let $\varepsilon > 0$ be such that

$$\int \log(f_x'(0) + \varepsilon)\, d\mu(x) < 0.$$

Let $a(x) = \log(f_x'(0) + \varepsilon)$. By the Birkhoff ergodic theorem, for μ-almost every $x \in X$

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} a(g^k(x)) = \int a(x)\, d\mu(x) < 0,$$

which implies $\lim_{n\to\infty} \sum_{k=0}^{n-1} a(g^k(x)) = -\infty$. This means that the function

$$A(x) = \max_{n\geq 1} \sum_{k=0}^{n-1} a(g^k(x)) \geq 0$$

is well-defined in almost every x.

By equicontinuity, there is $\delta > 0$ such that $0 \leq y \leq \delta$ implies $f_x'(x) \leq f_x'(0) + \varepsilon$ for all $x \in X$. Thus, if $y < \delta e^{-A(x)}$, we claim that

$$f_x^n(y) \leq \exp\left(\sum_{k=0}^{n-1} a(g^k(x)) - A(x)\right)\delta \leq \delta. \tag{2.4}$$

is true by induction. Indeed, the $n = 0$ case follows from the hypothesis on y, and if (2.4) is true for n, then by the mean value theorem,

$$\begin{aligned} f_x^{n+1}(y) &\leq \left(\max f'_{g^n(x)}([0, f_x^n(y)])\right) f_x^n(y) \\ &\leq (f'_{g^n(x)}(0) + \varepsilon) f_x^n(y) \\ &\leq \exp\left(a(g^n(x)) + \sum_{k=0}^{n-1} a(g^k(x)) - A(x)\right)\delta. \end{aligned}$$

Since $\lim_{n\to\infty} \sum_{k=0}^{n-1} a(g^k(x)) = -\infty$, we have $f_x^{n+1}(y) \to 0$. Thus, if $x \in \mathcal{B}(\mu)$ (which has full measure), then $e_n(x,y) \xrightarrow{w} \nu_0$ whenever $y \leq \delta e^{-A(x)}$.

□

Corollary 9. *If $(x,y) \mapsto \mathcal{S}f_x(y)$ is continuous and $\mathcal{S}f_x < 0$ for μ-almost every $x \in X$, then* $\mathrm{Lyap}(\mathcal{A}_0) + \mathrm{Lyap}(\mathcal{A}_1) < 0$*. In particular, at least one of the basins has positive measure.*

Proof. If suffices to integrate the logarithm of the inequality $f_x'(0)f_x'(1) < 1$ from lemma 6, and then apply lemma 8. □

2.4 Intermingled basins

The basins of the examples we saw before are continuity sets for the volume measure — that is, they are open sets modulo a zero volume set — and the union of all basins have full Lebesgue measure. But should basins of attractors constitute well-behaved sets in general? In some cases, such as for uniformly hyperbolic diffeomorphisms, this indeed the case, and there is a finite number of physical measures whose basins are continuity sets, and their union has full volume [6]. Outside the realm uniform hyperbolicity, though, this may not be the case — Kan [7] gave the first example of a partially hyperbolic endomorphism on a surface with two physical measures whose basins are intermingled, in the sense that for every open set U, the intersection of this set with either basin has positive volume [6]. Clearly, this implies both basins are dense in the surface, and therefore not continuity sets.

Refer to discussion in [8]

In this section we shall describe the construction of that flavor of example in a more general setting of skew products that also encompasses the original example. As before, let X be a compact metric space, $g\colon X \to X$ be ergodic for $\mu \in \mathcal{M}_1(X)$ and $F\colon X \times I \to X \times I$ a continous skew-product endomorphism of the form $F(x, y) = (g(x), f_x(y))$. We shall understand by "volume" the reference measure $m = \mu \otimes \mathrm{Leb}$. As mentioned in lemma 8, under some mild hypothesis for F, if the transverse Lyapunov exponents at the invariant boundaries $\mathcal{A}_i = X \times \{i\}$, given by

$$\mathrm{Lyap}(\mathcal{A}_i) = \int \log f_x'(i)\ d\mu(x)$$

for $i = 0, 1$ are both negative, then both basins have positive volume. Assuming this, the main property that will produce the intermingled phenomenon is the following hypothesis.

Hypothesis 1. There exists two periodic points x^+ and x^- in X, of periods n^+, n^-, such that for every $y \in (0, 1)$, $f_{x^+}^{n^+}(y) > y$ and $f_{x^-}^{n^-}(y) < y$.

Note that if we define the measures $\mu_i(A) = \mathrm{Leb}(A \cap \mathcal{B}_i)$, then the basins are intermingled if, and only if, $\operatorname{supp} \mu_i = X \times I$ for both $i = 0, 1$. We will show this is the case for $\operatorname{supp} \mu_0$, as the proof for $i = 1$ follows by symmetry. First, we shall collect some lemmas.

Lemma 10. *Given $x \in X$, $\operatorname{supp} \mu_0 \cap \{x\} \times I$ is a segment $\{x\} \times [0, a_x]$.*

Proof. By lemma 7, $\mathcal{B}_0$ is of the above form. Since we know that the support of a measure is closed, we can take $y < 1$ such that $(x, y) \in \operatorname{supp} \mu_0$ and show that if $z < y$ then $(x, z) \in \operatorname{supp} \mu_0$. Let $U = B_\varepsilon(x, z)$ be a ball centered at (x, z) of radius smaller than $1 - y$, and $V = B_\varepsilon(x, y)$. Note that

$$V \cap \mathcal{B}_0 \subseteq \{(u, v + y - z) \mid (u, v) \in U \cap \mathcal{B}_0\},$$

so since $(x, y) \in \operatorname{supp} \mu_0$, and Leb is invariant by vertical translations, $\mathrm{Leb}(U \cap \mathcal{B}_0) \geq \mathrm{Leb}(V \cap \mathcal{B}_0) > 0$. □

Lemma 11. *The support $\operatorname{supp} \mu_0$ is an invariant set, that is, $F^{-1}(\operatorname{supp} \mu_0) = \operatorname{supp} \mu_0$.*

Proof. If $p \in F^{-1}(\operatorname{supp}\mu_0)$ and U is a neighborhood of p, then since F is an open map, $F(U)$ is a neighborhood of $F(p) \in \operatorname{supp}\mu_0$, so $\mu_0(F(U)) > 0$. Since F is Lipschitz, if $\mu_0(U)$ were 0, then $\mu_0(F(U)) = \operatorname{Leb}(F(U) \cap \mathcal{B}_0) = \operatorname{Leb}(F(U \cap \mathcal{B}_0)) = 0$, which is not the case, so $\mu_0(U) > 0$. □

Corollary 12. $\operatorname{supp}\mu_0 = X \times I$.

Proof. Since $\operatorname{supp}\mu_0$ is closed and by hypothesis, $F^n(x^-, y) \to (x^-, 0)$ for all $y \in (0,1)$, then $\{0\} \times I \subseteq \operatorname{supp}\mu_0$. □

Proof. By symmetry, it is enough to show $\mathcal{A}_1 \subseteq K$. Let $(x, y) \in K$ be the point in the statement.

Inductively, we can construct a backward orbit $\cdots \mapsto x_{-2} \mapsto x_{-1} \mapsto x_0$ of g such that $x_{-n-1} \in g^{-1}(\{x_{-n}\})$ is the closest preimage of x_-. Since x_- is a repelling fixed point, the construction implies that $x_{-n} \to x_-$, so by **??** 1, for every $y \in (0,1)$ there exists n_0 such that $n \geq n_0$ implies $f_{x_{-n}}^{-1}(y_0) > y$. In particular,

$$y_{-n} = f_{x_{-n}}^{-1} \circ \cdots \circ f_{x_{-1}}^{-1}(y_0)$$

is such that $y_{-n} \to 1$. Since $\operatorname{supp}\mu_0$ is closed and $(x_{-n}, y_{-n}) \in F^{-n}(\operatorname{supp}\mu_0)$, we conclude that $(x_-, 1) \in \operatorname{supp}\mu_0$. Now, notice that $E = \bigcup_{n=0}^{\infty} g^{-1}(\{x_-\})$ is dense in X and 1 is a fixed point of the f_x. Thus, $E \times \{1\} \subseteq \operatorname{supp}\mu_0$ and consequently $\mathcal{A}_1 \subseteq \operatorname{supp}\mu_0$. □

Corollary 13. *Suppose* **??** *1 holds and both basins have positive measure. Then the basins $\mathcal{B}_0$ and $\mathcal{B}_1$ are intermingled for the Lebesgue measure on the cylinder.*

Proof. By definition, the basins are strictly invariant sets. If we define the measures $\mu_i(A) = \operatorname{Leb}(A \cap \mathcal{B}_i)$, then the basins are intermingled if, and only if, $\operatorname{supp}\mu_i = X \times I$ for both $i = 0, 1$. We will show this is the case for $\operatorname{supp}\mu_0$, as the proof for $i = 1$ is analogous.

Claim1: The support $\operatorname{supp}\mu_0$ is backwards-invariant.

Indeed, if $p \in F^{-1}(\operatorname{supp}\mu_0)$ and U is a neighborhood of p, then since F is an open map, $F(U)$ is a neighborhood of $F(p) \in \operatorname{supp}\mu_0$, so $\mu_0(F(U)) > 0$. Since F is Lipschitz, if $\mu_0(U)$ were 0, then $\mu_0(F(U)) = \operatorname{Leb}(F(U) \cap \mathcal{B}_0) = \operatorname{Leb}(F(U \cap \mathcal{B}_0)) = 0$, which is not the case, so $\mu_0(U) > 0$.

Claim2: There exists a point $(x_0, y_0) \in \operatorname{supp}\mu_0 \setminus X \times \{0\}$.

Indeed, $\mu_0(\operatorname{supp}\mu_0) \geq \operatorname{Leb}(\mathcal{B}_0) > 0$, so $\mu_0(\operatorname{supp}\mu_0 \setminus X \times \{0\}) > 0$ and the set cannot be empty.

Claim3: Given $x \in X$, $\operatorname{supp}\mu_0 \cap \{x\} \times I$ is a segment $\{x\} \times [0, a_x]$.

By lemma 7, $\mathcal{B}_0$ is of the above form. To show that the support also has this property, we can take $y < 1$ such that $(x, y) \in \operatorname{supp}\mu_0$ and show that if $z < y$ then $(x, z) \in \operatorname{supp}\mu_0$. Let $U = B_\varepsilon(x, z)$ be a ball centered at (x, z) of radius smaller than $1 - y$, and $V = B_\varepsilon(x, y)$. Note that $V \cap \mathcal{B}_0 \subseteq (U \cap \mathcal{B}_0) + (0, y - z)$, and since $(x, y) \in \operatorname{supp}\mu_0$, $\operatorname{Leb}(U \cap \mathcal{B}_0) \geq \operatorname{Leb}(V \cap \mathcal{B}_0) > 0$. This shows the claim.

Now, by the first two claims and **??**, $\operatorname{supp}\mu_0$ contains both boundaries. By the last claim, it is composed of vertical segments, so it must be equal to $X \times I$. □

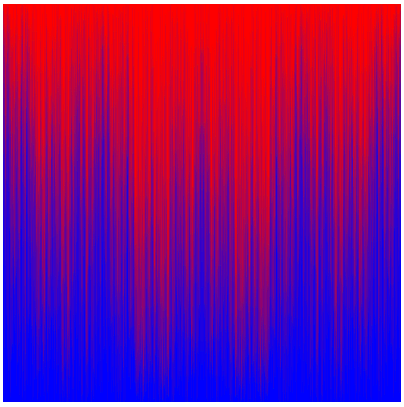

Figure 2.1: Picture of the basins of example 2.4.1. It should not be taken too seriously, as it is evidently not possible to draw intermingled sets.

Example 2.4.1. Define $q_a(y) = y + ay(1-y)$ and $p(x) = \varepsilon\cos(2\pi x)$ for some $0 < \varepsilon < 1$. If $|a| < 1$, then q_a is a diffeomorphism of the interval I. If $F(x) = (3x, q_{p(x)}(y))$, then $x^+ = 0$ and $x^- = 1/2$ are fixed points of the map $g(x) = 3x$ (where we identify S^1 with $[0,1]$) that satisfy **??** 1, because $q_a(y) > y$ if $a > 0$ and $q_a(y) < y$ if $a < 0$ for all $y \in (0,1)$. Moreover, since q_a is a degree two polynomial, it has negative Schwarzian whenever $a \neq 0$, and from the symmetry of p we see both basins of $F(x,y) = (kx, q_{p(x)}(y))$ must have positive measure, because at least one of them has. So F displays intermingled basins.

2.5 Negative Schwarzian

The statement of the theorem below corresponds to [4, Theorem 3.2] without the hypothesis that both basins have positive measure. The question of whether this hypothesis could be removed was left open in the article, and the following proof shows that it is indeed unnecessary.

Theorem 14. *If $(x,y) \mapsto \mathcal{S}f_x(y)$ is continuous and $\mathcal{S}f_x < 0$ almost everywhere in $x \in X$, then there exists a measurable function $\sigma\colon X \to I$ defined almost everywhere such that*

$$(x,y) \in \mathcal{B}_0 \text{ whenever } y < \sigma(x) \quad \text{and} \quad (x,y) \in \mathcal{B}_1 \text{ whenever } y > \sigma(x).$$

Consequently, the union $\mathcal{B}_0 \cup \mathcal{B}_1$ has total $\mu \otimes \mathrm{Leb}$ measure.

Proof. We start by defining functions $\xi_0, \xi_1 : X \to I$ that separate the basins in the following sense:

$$\xi_0(x) = \sup\{y \in I : y \in \mathcal{B}_0\}, \qquad \xi_1(x) = \inf\{y \in I : y \in \mathcal{B}_1\}.$$

By lemma 7, if $y < \xi_0(x)$ or $y > \xi_1(x)$, the empirical measures $e_n(x,y)$ converge to $\nu_0 = \mu \otimes \delta_0$ or to $\nu_1 = \mu \otimes \delta_1$, and if $\xi_0(x) < y < \xi_1(x)$, they do not converge to either ν_0 or ν_1.

It is sufficient to show that $A = \{x \in X : \xi_0(x) \neq \xi_1(x)\}$ has measure zero, since in this case we can define σ as the common value of the two functions. Suppose by contradiction that this set does not have measure zero. Since the f_x are bijective, A is an invariant set under g, so the ergodicity of μ implies that A has full measure.

Since we are under the hypotheses of corollary 9, we can assume without loss of generality that $\mathrm{Leb}(\mathcal{A}_1) > 0$. By ergodicity, we also have $\xi_1(x) < 1$ for μ-almost every $x \in X$. Thus, given $k \in \mathbb{N}$, there exists $\delta_k < 1$ such that $E_k = \{x \in X : \xi_1(x) < \delta_k\}$ has measure $\mu(E_k) > 1 - 1/k$. Applying the Birkhoff ergodic theorem to the functions 1_{E_k}, we obtain $B \subseteq X$ of full measure such that for all $x \in B$ and k,

$$\lim_{n\to\infty} \frac{1}{n}\#\{0 \le i < n \mid \xi_1(g^i(x)) < \delta_k\} = \mu(E_k) > 1 - 1/k. \tag{2.5}$$

By the ergodicity of μ, there exists another set $C \subseteq X$ of full measure whose empirical measures converge to μ, and since $\mathcal{S}f_x \prec 0$ almost everywhere in $x \in X$, there exists a set D of full measure such that $\mathcal{S}f_{g^n(x)} \prec 0$ for all $n \ge 0, x \in D$. Let $x_0 \in A \cap B \cap C \cap D$ and $y_0 \in (\xi_0(x), \xi_1(x))$, and let $\nu \in \mathcal{M}(x, y) \setminus \{\nu_0\}$ be a limit point of the empirical measures different from ν_0. Note that since $x_0 \in B$, necessarily $(\pi_X)_*\nu = \mu$, but since $\nu \neq \nu_0$, it follows that $\nu(X \times \{0\}) < 1$. Thus, $\operatorname{supp} \nu$ is not contained in $X \times \{0\}$, and there exists $\varepsilon > 0$ such that $U = X \times (\varepsilon, 1]$ satisfies $\nu(U) > 0$. Let k be such that $\nu(U) > 1/k$, and $n_j \to \infty$ such that $\liminf_{j\to\infty} e_{n_j}(x_0, y_0)(U) > 1/k$. In other words, we have

$$\liminf_{j\to\infty} \frac{1}{n_j}\#\{0 \le i < n_j \mid f^i_{x_0}(y_0) > \varepsilon\} > 1/k. \tag{2.6}$$

Now, consider the sequence $r_n = \rho(0, f^n_{x_0}(y_0), f^n_{x_0}(\xi_1(x_0)), 1)$, which is well-defined and positive since $0 < f^n_{x_0}(y_0) < f^n_{x_0}(\xi_1(x_0)) < 1$ by the definition of the points. Note also that the graph of ξ_1 is strictly invariant under F, so $f^n_{x_0}(\xi_1(x_0)) = \xi_1(g^n(x_0))$.

Since $x \in D$, it follows by lemma 5 that r_n is an increasing sequence. On one hand, it is not possible for $r_n \to \infty$, because from eqs. (2.5) and (2.6) we know that there exist $n \in \mathbb{N}$ arbitrarily large such that $0 < \varepsilon < f^n_{x_0}(y_0) < f^n_{x_0}(\xi_1(x_0)) < \delta_k < 1$. On the other hand, if $r_n \to r < \infty$, then for each $x \in X$ there exists a unique $\xi(x)$ such that $\rho(0, \xi(x), \xi_1(x), 1) = r$. In fact, a calculation shows that

$$\xi(x) = \frac{\xi_1(x)}{r(1 - \xi_1(x)) + \xi_1(x)}.$$

Note that $r_n \to r$ implies $\lim_{n\to\infty}|\xi(g^n(x_0)) - f^n_{x_0}(y_0)| = 0$. Since we know that x_0 is in the basin of μ, necessarily $e_n(x_0, y_0) \xrightarrow{w} \eta$, where $\eta = (\operatorname{graph} \xi)_*\mu$ and $(\operatorname{graph} \xi)(x) = (x, \xi(x))$. Thus, η is an invariant measure. But then,

$$\varphi(x, y) = \frac{\rho(0, y, \xi_1(x), 1)}{1 + \rho(0, y, \xi_1(x), 1)}$$

is an integrable function defined at η-almost every point, with the property that $\varphi(F(x, y)) > \varphi(x, y)$ at η-almost every point. This is impossible, as it would imply $\int \varphi \, d\eta < \int \varphi \circ F \, d\eta = \int \varphi \, d\eta$. □

Example 2.5.1. In example 2.4.1, we have $q_a'(y) = 1 + a(1 - 2y)$, $q_a''(y) = -2a$ and $q_a''' \equiv 0$, so $\mathcal{S}q_a(y) < 0$ whenever $a \neq 0$. Since $q(x) = 0$ only at two points, by theorem 14 there exists $\sigma\colon S^1 \to I$ that separates the two basins.

2.6 Positive Schwarzian

If $f\colon M \to M$ is a diffeomorphism on a manifold, we say that an invariant measure μ is *asymptotic* with respect to a reference measure if the empirical measures of almost every point of the reference measure converges to μ. Intuitively, μ describes the entire statistical behavior of the system, up to a set of zero volume.

Since theorem 14 does not require that both basins have positive measure, we also obtain a more general statement than the version in [4, Theorem 5.1], which requires both Lyapunov exponents to be positive. The proof below is also somewhat different, more similar to the proof of theorem 14.

Theorem 15. *If $\mathcal{S}f_x > 0$ almost everywhere, then F has an asymptotic measure ν with respect to $\mu \otimes \mathrm{Leb}$. Furthermore, for each $i \in \{0,1\}$, $\mathrm{Lyap}(\mathcal{A}_i) > 0$ implies $\nu(\mathcal{A}_i) = 0$.*

The idea of the proof is to pass to the invertible extension of the system and observe that the inverses of the fiber maps have negative Schwarzian. By applying theorem 14, we obtain an invariant graph that separates the two basins of the inverse transformation. Since the orbits diverge from the graph for the inverse of the extension, they converge to the graph for the extension. Thus, the graph describes an asymptotic measure for the extension. The pushforward of this measure is the asymptotic measure for the original map.

Proof. Let $(\tilde{X}, \tilde{g}, \tilde{\mu})$ be the invertible extension of (X, g, μ), $\pi\colon (\tilde{X}, \tilde{g}, \tilde{\mu}) \to (X, g, \mu)$ the associated conjugation, and $\tilde{F}\colon \tilde{X} \times I \to \tilde{X} \times I$ given by

$$\tilde{F}(\tilde{x}, y) = (\tilde{g}(\tilde{x}), f_{\pi(\tilde{x})}(y)).$$

Note that $\tilde{F}^{-1}(x, y) = (\tilde{g}^{-1}(x), f^{-1}_{\pi(\tilde{g}^{-1}(x))}(y))$. Since $\pi_*\tilde{\mu} = \mu$ and $\tilde{g}_*\tilde{\mu} = \tilde{\mu}$, by proposition 3 we have $f^{-1}_{\pi(\tilde{g}^{-1}(\tilde{x}))} < 0$ for almost every $\tilde{x} \in \tilde{X}$. Thus, we can apply theorem 14 to F^{-1} and obtain a function $\sigma\colon \tilde{X} \to I$ that separates the basins $\tilde{\mathcal{B}}_1$ and $\tilde{\mathcal{B}}_0$.

The hyperbolic metric (which coincides with the Hilbert metric) on $(0,1)$ is given by

$$d_H(r, s) = \log|\rho(0, r, s, 1)|.$$

This metric induces the usual topology of the interval. Since maps with positive Schwarzian decrease cross-ratios, the condition $\mathcal{S}f_x > 0$ means that f_x is a contraction (not necessarily uniform) for this metric.

Let $A \subseteq \tilde{X}$ be a set of full $\tilde{\mu}$ measure where $\mathcal{S}f_{\pi(\tilde{g}^n(\tilde{x}))} > 0$ for all $n \geq 0, \tilde{x} \in A$, and $B \subseteq \tilde{X}$ be another set of full measure where the empirical measures of $\tilde{x} \in B$ converge to the ergodic measure $\tilde{\mu}$. It follows that for all $\tilde{x} \in A \cap B$ and $y \in (0,1)$, the limit $r = \lim_{n\to\infty} d_H(f^n_{\pi(\tilde{x})}(y), \sigma(\tilde{g}^n(\tilde{x})))$ exists and is zero if and only if the orbit of $(\tilde{x}, y)$ converges in distance to the graph of σ under $\tilde{F}$, which in turn implies that $e_n(\tilde{x}, y) \xrightarrow{w} \tilde{\nu} = (\mathrm{graph}\,\sigma)_*\tilde{\mu}$.

Suppose $r > 0$. Then, as in the previous proof, there would exist another graph ξ, different from σ almost everywhere, such that $e_n(\tilde{x}, y) \xrightarrow{w} \eta = (\mathrm{graph}\,\xi)_*\tilde{\mu}$. Since $\varphi(\tilde{x}, y) = d_H(y, \sigma(\tilde{x}))$

is such that $\varphi(F(\tilde{x}, y)) < \varphi(\tilde{x}, y)$ whenever $(\tilde{x}, y) \in \operatorname{graph} \xi$ and η is an invariant measure for F with $\eta(\operatorname{graph} \xi) = 1$, we reach a contradiction.

This shows that $E = (A \cap B) \times (0, 1)$ is a set of full $\tilde{\mu} \otimes \mathrm{Leb}$ measure such that $(\tilde{x}, y) \in E$ implies $e_n(\tilde{x}, y) \xrightarrow{w} \tilde{\nu}$. Let $\nu = \pi_* \tilde{\nu} \in \mathcal{M}(X \times I)$. The measure ν is invariant, since

$$F_* \nu = (F \circ \pi)_* \tilde{\nu} = (\pi \circ \tilde{F})_* \tilde{\nu} = \pi_* \tilde{\nu} = \nu.$$

Furthermore, the basin of ν has full Lebesgue measure. In fact, if $\varphi \in C(X \times I)$, then $\varphi \circ \pi \in C(\tilde{X} \times I)$ and $e_n(\pi(\tilde{x}), y)(\varphi) = e_n(\tilde{x}, y)(\varphi \circ \pi)$. Therefore, $e_n(x, y) \to \nu$ if and only if $e_n(\tilde{x}, y) \to \tilde{\nu}$ for any $\tilde{x} \in \pi^{-1}(x)$, that is, $(\pi \times \mathrm{id})^{-1}(\mathcal{B}(\nu)) = \mathcal{B}(\tilde{\nu})$. It follows that $(\mu \otimes \mathrm{Leb})(\mathcal{B}(\nu)) = (\tilde{\mu} \otimes \mathrm{Leb})(\mathcal{B}(\tilde{\nu})) = 1$. □

Chapter 3

Historical behavior

Ergodic theory tries to find order in chaos by hoping that even if a dynamical system looks complicated, its statistical properties are well behaved. But what if the statistics don't even exist for most points? This phenomenon is what is called *non-statistical*, or *historical* behavior.

3.1 Martingale-coboundary decomposition

The key idea for applying theorem 30 in a dynamical system $(X, \mathcal{F}, \mu, f)$, where μ is a f-invariant probability, is finding a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ for which $\varphi \in L^2(\mu)$ can be decomposed as a sum

$$\varphi = \varphi_0 + \phi \circ f - \phi \tag{3.1}$$

such that $\phi \in L^2(\mu)$, φ_0 is $\mathcal{G}$-measurable and $\eta = (\varphi_0 \circ f^{i-1})_{i \geq 1}$ is a reversed martingale difference sequence. Indeed, this allows the transference of the weak invariant principle for the original φ: let $\xi = (\varphi \circ f^{i-1})_{i \geq 1}$, and notice that

$$S_n^{\xi} = S_n^{\eta} + \phi \circ f^n - \phi,$$

so

$$D_n^{\xi}(x)(t) = \frac{S_{\lfloor nt \rfloor}^{\xi}(x)}{\sqrt{n}} = D_n^{\eta}(x)(t) + \frac{1}{\sqrt{n}}\big(\phi \circ f^{\lfloor nt \rfloor} - \phi\big)(x).$$

It suffices to show that for μ-almost every x, the second term in the sum converges uniformly to zero (as a function of t). Given $\varepsilon > 0$, we look at the sets $B_n = \{x : (\phi^2 \circ f^n)(x) \geq n\varepsilon\}$. Since

$$\sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(\{\phi^2 \geq n\varepsilon\}) = \int \lfloor \phi^2/\varepsilon \rfloor \, d\mu \leq \frac{1}{\varepsilon}\|\phi\|_2^2,$$

by Borel-Cantelli, $\mu(\limsup_{n\to\infty} B_n) = 0$. If we take the intersection over k of the sets $\liminf_{n\to\infty} B_n^c$ for $\varepsilon = 1/k$, we get a full-measure set G with the property that for every $x \in G$ and $\varepsilon > 0$, there exists N such that $n \geq N$ implies $\frac{1}{\sqrt{n}}|\phi(f^n(x))| \leq \varepsilon$. Now, let $M \geq N$ be big enough so that $n \geq M$ implies $\frac{1}{\sqrt{n}}|\phi(f^i(x))| \leq \varepsilon$ for all $0 \leq i \leq N$. So for $n \geq M$,

$$\sup_{t \in [0,1]} \frac{1}{\sqrt{n}}\big|\phi \circ f^{\lfloor nt \rfloor} - \phi\big|(x) \leq \max_{0 \leq i \leq N} \frac{|\phi \circ f^i(x)|}{\sqrt{n}} + \max_{N \leq i \leq n} \frac{|\phi \circ f^i(x)|}{\sqrt{i}} \leq 2\varepsilon$$

as desired.

Existence of martingale-coboundary decomposition

From now on, we assume that μ is an ergodic measure for f.

Theorem 16. *Suppose $\varphi \in L^2(\mu)$ and $\mathcal{F}_j = f^{-j}(\mathcal{F})$ are such that*

$$\sum_{n=0}^{\infty} \|\mathbb{E}_\mu(\varphi \mid \mathcal{F}_j)\|_2 < \infty.$$

Then a decomposition as in eq. (3.1) *exists. Moreover, this implies that*

$$\int \varphi \, d\mu = \int \varphi_0 \, d\mu = 0$$

and that the series in

$$\sigma^2 = \|\varphi_0\|_2^2 = \|\varphi\|_2^2 + 2 \sum_{n=1}^{\infty} \int \varphi \cdot \varphi \circ f^n \, d\mu$$

converges absolutely. By the preceding arguments, if $\varphi_0 \neq 0$ then $(D_n^\xi)_* \mu \xrightarrow{w} \sigma W$.

Proof. We begin by writing $\phi = \phi^- + \phi^+$, where $\phi^+ = \mathbb{E}_\mu(\phi \mid \mathcal{G}_0)$.

write □

Remark 3. If φ was a coboundary $\varphi = \phi \circ f - \phi$ to begin with, then it is clear that $S_n^\xi = \phi \circ f^n - \phi$, so the iterates of the skew product $F(x, y) = (f(x), y + \varphi(x))$ satisfy

$$F^n(x_0, y_0) = (f^n(x_0), \phi(f^n(x_0)) - \phi(x_0) + y_0) \subseteq \operatorname{graph}(\psi)$$

and are thus contained in the graph of ψ, where $\psi(x) = \phi(x) - \phi(x_0) + y_0$. In particular, F is not ergodic for $\mu \otimes \mathrm{Leb}$.

Corollary 17. *Suppose φ is such that there exists a summable sequence $c_n > 0$ satisfying*

$$\int (\psi \circ f^n)\varphi \, d\mu \leq \|\psi\|_2 c_n$$

for all $\psi \in L^2(\mathcal{G})$, where $\mathcal{G}$ is as in theorem 16. Then

$$\sum_{n=0}^{\infty} \|\mathbb{E}(\varphi \mid f^{-n}(\mathcal{G}))\|_2 < \infty.$$

Remark 4. The book [9] proves this condition is satisfied for smooth expanding maps, Hölder-continuous φ and $\mathcal{G} = \mathcal{F}$, so this includes the $f(x) = kx$ maps. The second condition is trivial when $\mathcal{G} = \mathcal{F}$.

Remark 5. More generally, [9] also shows the decomposition (3.1) exists for uniformly hyperbolic f and Hölder φ, also by showing that a sum like in theorem 16 converges due to decay of correlations, then uses this to show the central limit theorem (but I have not read the proof). In this case, $\mathcal{G}$ has something to do with the attractor and the second condition follows from Hölder somehow.

the examples in the Billingsley book are worth checking. They include both the kx maps (hmm... I guess B-M didn't see it?) and the Gauss map.

3.2 Historical behavior for skew products

Let (X, μ) be a Borel probability space, where X is compact and metrizable, and $F\colon X \times I \to X \times I$ a skew product of the form

$$F(x, y) = (g(x), f_x(y))$$

where $g\colon X \to X$ is ergodic and measure-preserving. Suppose there is a homeomorphism $\gamma\colon I \to \overline{\mathbb{R}}$ and a measurable function $p\colon X \to \mathbb{R}$ satisfying $t_x \circ \gamma = \gamma \circ f_x$ for all $x \in X$, where $t_x(y) = y + p(x)$ is the translation by $p(x)$.

$$\begin{tikzcd} I \arrow[r, "f_x"] \arrow[d, "\gamma"] & I \arrow[d, "\gamma"] \\ \overline{\mathbb{R}} \arrow[r, "t_x"] & \overline{\mathbb{R}} \end{tikzcd}$$

Note there is no loss of generality in assuming $\gamma(0) = -\infty$, $\gamma(1) = +\infty$ and $\gamma\left(\frac{1}{2}\right) = 0$.

Theorem 18. *If the weak invariance principle is valid for the pair (g, p), in the sense $(D_n^{\xi})_*\mu \xrightarrow{w} \sigma W$ for $\sigma > 0$ and $\xi = (p \circ g^{i-1})_{i \geq 1}$, then F has historical behavior. In fact, for μ-almost every $x \in X$ and every $y \in (0, 1)$,*

$$\{e_n(x, y)\}'_{n \in \mathbb{N}} = [\delta_-, \delta_+] \tag{3.2}$$

where $\delta_- = \mu \otimes \delta_0$ and $\delta_+ = \mu \otimes \delta_1$ and the limit points are taken in the weak topology.*

Proof. Fix $\delta \in (0, 1)$, and define

$$\begin{aligned} T(x, y) &= \limsup_{n \to \infty} \frac{1}{n} \#\{0 \leq i < n \mid f_x^{(i)}(y) > \delta\} \\ &= \limsup_{n \to \infty} \frac{1}{n} \sum_{i=0}^{n-1} 1_{(M_y, \infty)}(S_i^{\xi}(x)) \end{aligned}$$

where $M_y = \gamma(\delta) - \gamma(y)$. Let E be the full μ-measure set given by lemma 28.

Claim1: $T(x, y_1) = T(x, y_2)$ for all $x \in E$, $y_1, y_2 \in (0, 1)$.

Indeed, set $M = \max\{|M_{y_1}|, |M_{y_2}|\}$ and let $G = \{i : |S_i^{\xi}(x)| > M\}$. Since $1_{(M_{y_1}, \infty)}(t) = 1_{(M_{y_2}, \infty)}(t)$ whenever $|t| > M$, for all $i \in G$,

$$1_{(M_{y_1}, \infty)}(S_i^{\xi}(x)) = 1_{(M_{y_2}, \infty)}(S_i^{\xi}(x)).$$

So

$$\begin{aligned} |T(x, y_1) - T(x, y_2)| &\leq \limsup_{n \to \infty} \frac{1}{n} \sum_{i=0}^{n-1} |1_{(M_{y_1}, \infty)}(S_i^{\xi}(x)) - 1_{(M_{y_2}, \infty)}(S_i^{\xi}(x))| \\ &\leq \limsup_{n \to \infty} \frac{1}{n} \#\{0 \leq i < n \mid i \notin G\} = 0. \end{aligned}$$

This proves the claim. Since T is invariant by F, for any $x \in g^{-1}(E)$,

$$T(x, y) = T(g(x), f_x(y)) = T(g(x), y).$$

By ergodicity, there is a full measure set $H \subseteq X$ and $t \in [0,1]$ for which $T(x,y) = t$ for all $x \in H$, $y \in (0,1)$.

We can use the arcsine law to evaluate this constant. Given $\varepsilon > 0$, fix some $y \in (0,1)$ and define

$$E_n = \left\{ x \in X \;\middle|\; \frac{1}{n}\#\{0 \le i < n \mid S_i^{\xi}(x) > M_y\} > 1 - \varepsilon \right\}.$$

By the lemma 27, we know $\lim_n \mu(E_n) = Z([1-\varepsilon, 1]) > 0$, so

$$\mu(\limsup_{n\to\infty} E_n) \ge \limsup_{n\to\infty} \mu(E_n) > 0.$$

This means there exists $x_0 \in H \cap \limsup E_n$, and since

$$\limsup_{n\to\infty} E_n = \{x \in X \mid T(x,y) > 1 - \varepsilon\}$$

we have $t = T(x_0, y) > 1 - \varepsilon$. But $\varepsilon > 0$ is arbitrary, so $T(x,y) = 1$ for every $x \in H$ and $y \in (0,1)$.

So far, we started with $\delta \in (0,1)$ and showed that for

$$T_\delta(x,y) = \limsup_{n\to\infty} \frac{1}{n}\#\{0 \le i < n \mid f_x^{(i)}(y) > \delta\},$$

there is $H_\delta \subseteq X$ with $\mu(H_\delta) = 1$ such that for $x \in H_\delta$, $y \in (0,1)$, $H_\delta(x,y) = 1$. But note that for fixed x, y, the function $\delta \mapsto T_\delta(x,y)$ is monotone. With this observation, it's easy to see that if we define

$$H = \bigcap_{i=1}^{\infty} H_{1/i} \cap H_{1-1/i},$$

then $\mu(F) = 1$ and $T_\delta(x,y) = 1$ for all $x \in H$ and $y, \delta \in (0,1)$.

In particular, this implies that if $x \in H$ and $y \in (0,1)$, then for any $\delta \in (0,1)$, letting $U_n = \{0 \le i < n \mid f_x^{(i)}(y) > \delta\}$, the limit supremum of the y coordinate satisfies

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} f_x^{(i)}(y) \ge \limsup_{n\to\infty} \frac{1}{n} \sum_{i \in U_n} \delta = \delta,$$

so the left-hand side is equal to 1. The proof we have so far works analogously for the time averages of the iterates in which $f_x^{(i)}(y) < \delta$, and by doing so we get

$$\liminf_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} f_x^{(i)}(y) = 0.$$

Note that those are time averages of the continuous bounded function $\pi_2 \colon X \times I \to \mathbb{R}$, so historical behavior is proved.

We now focus on eq. (3.2). By Birkhoff, we can fix a full measure set $S \subseteq X$ such that for all $x \in S$ and $\varphi \in C(X)$,

$$\frac{1}{n} \sum_{i=0}^{n-1} \varphi(g^i(x)) \to \int_X \varphi \, d\mu.$$

Given x in the full measure set $H \cap S$ and $y \in (0,1)$, we construct an increasing sequence $(n_k^-)_k \subseteq \mathbb{N}$ inductively, letting $n_0^- = 1$ and $n_{k+1}^- > n_k^+$ be any integer satisfying

$$\frac{1}{n_k^-}\#\{0 \le i < n_k^- \mid f_x^{(i)}(y) < 1/k\} > 1 - 1/k.$$

Analogously, we can pick an increasing sequence $(n_k^+)_k \subseteq \mathbb{N}$ for which

$$\frac{1}{n_k^+}\#\{0 \le i < n_k^+ \mid f_x^{(i)}(y) > 1 - 1/k\} > 1 - 1/k.$$

Claim2:

$$e_{n_k^-}(x,y) \xrightarrow{w} \delta_- \quad \text{and} \quad e_{n_k^+}(x,y) \xrightarrow{w} \delta_+.$$

We only show the first convergence, since the second is analogous.

Let $\phi \in C(X \times I)$ and $\varepsilon > 0$. Since ϕ is uniformly continuous, there is some k_1 for which

$$\sup\big\{|\phi(x,1) - \phi(x,s)| : x \in X, s \in [0, 1/k_1]\big\} < \varepsilon.$$

Let $D = \{i \mid f_x^{(i)}(y) < 1/k_1\}$, and let N be big enough so that for $k \ge N$,

$$\left|\frac{1}{n_k^-}\sum_{i=0}^{n_k^- - 1}\phi(F^i(x,0)) - \int_X \phi(z,0)\, d\mu(z)\right| < \varepsilon$$

(which is possible since $x \in S$ and $\varphi(x) = \phi(x,0)$ is continuous), and

$$\frac{1}{n_k^-}\#\{0 \le i < n_k^- \mid i \notin D\} < \varepsilon/\|\phi\|_\infty,$$

so we can estimate

$$\begin{aligned}
&|e_{n_k^-}(x,y)(\phi) - \delta_-(\phi)| \\
&\qquad = \left|\frac{1}{n_k^-}\sum_{i=0}^{n_k^- - 1}\phi(F^i(x,y)) - \int_X \phi(z,0)\, d\mu(z)\right| \\
&\qquad \le \left|\frac{1}{n_k^-}\sum_{i=0}^{n_k^- - 1}\phi(F^i(x,y)) - \frac{1}{n_k^-}\sum_{i=0}^{n_k^- - 1}\phi(F^i(x,0))\right| + \varepsilon \\
&\qquad \le \frac{1}{n_k^-}\sum_{\substack{0 \le i < n_k^- \\ i \in D}}\big|\phi(F^i(x,y)) - \phi(F^i(x,0))\big| + \frac{2}{n_k^-}\sum_{\substack{0 \le i < n_k^- \\ i \notin D}}\|\phi\|_\infty + \varepsilon \\
&\qquad \le \frac{1}{n_k^-}\sum_{\substack{0 \le i < n_k^- \\ i \in D}}\big|\phi(F^i(x,y)) - \phi(F^i(x,0))\big| + 3\varepsilon \\
&\qquad \le \varepsilon + 3\varepsilon = 4\varepsilon.
\end{aligned}$$

This completes the proof of the claim. To finish the proof of (3.2), we first show the following.

Claim3: If $\eta \in \mathcal{M}(x,y)$, then $\operatorname{supp}\eta \subseteq X \times \{0,1\}$ and $\eta \in [\delta_0, \delta_1]$.

We use lemma 28 again for this. Let $e_{n_k}(x,y) \xrightarrow{w} \eta$ be a limit point, $x_0 \in X$ and $y_0 \in (0,1)$, so there is a neighborhood U of (x_0, y_0) such that $\overline{U} \subseteq X \times (0,1)$. By the lemma, $e_{n_k}(x,y)(U) \to 0$, so by Portmanteau

$$\eta(U) \leq \liminf_{k\to\infty} e_{n_k}(x,y)(U) = 0.$$

This shows $(x_0, y_0) \notin \operatorname{supp} \eta$.

We know η is a limiting measure for x, so it's invariant by F, and the fact $x \in X$ implies $(\pi_1)_*\eta = \mu$. This means that $\eta \ll \delta_- + \delta_+$, and since δ_-, δ_+ are ergodic measures, η must be a convex combination of the two.

We have shown $\mathcal{M}(x,y) \subseteq [\delta_-, \delta_+]$. To conclude eq. (3.2), it suffices to note that by proposition 1, $\mathcal{M}(x,y)$ is connected, and since it contains both δ_- and δ_+, it must be the whole segment. □

Remark 6 (A different proof). Another way to show $\delta_-, \delta_+ \in \mathcal{M}(x)$ for x in a full measure set H is as follows: let $\delta > 0$ and $p_\delta \colon [0,1] \to \mathbb{R}$ be any continuous function such that $p_\delta(0) = 0$ and $p(t) > 1 + \delta$ for $t > \delta$. Let $U_\delta = B_\delta(p_\delta)$. By the Forgery theorem we know $W(U_\delta) > 0$, so by

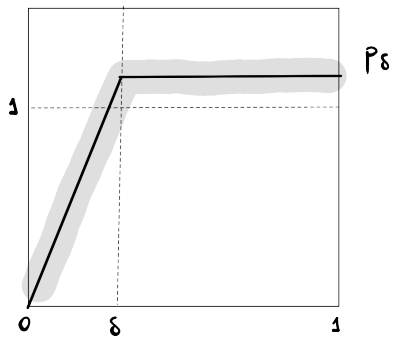

Figure 3.1: The function p_δ

theorem 21

$$\liminf_{n\to\infty}((D_n^\xi)_*\mu)(U_\delta) \geq \sigma W(U_\delta) > 0.$$

Therefore,

$$S_\delta = \limsup_{n\to\infty} (D_n^\xi)^{-1}(U_\delta) \subseteq X$$

has positive measure. This set is also invariant for g because

$$\begin{aligned}
g^{-1}(S_\delta) &= \limsup_{n\to\infty}\{x \in X \mid D_n^\xi(g(x)) \in U_\delta\} \\
&= \left\{ x \in X \;\middle|\; \liminf_{n\to\infty} \sup_t \left| \frac{S^\xi_{\lfloor nt \rfloor + 1}(x) - p(x)}{\sqrt{n}} - p_\delta(t) \right| < \delta \right\} \\
&= \left\{ x \in X \;\middle|\; \liminf_{n\to\infty} \sup_t \left| \frac{S^\xi_{\lfloor nt' \rfloor}(x)}{\sqrt{n}} - p_\delta(t) \right| < \delta \right\} \text{ where } t' = t + 1/n \\
&= \left\{ x \in X \;\middle|\; \liminf_{n\to\infty} \sup_t \left| \frac{S^\xi_{\lfloor nt \rfloor}(x)}{\sqrt{n}} - p_\delta(t) \right| < \delta \right\} = S_\delta.
\end{aligned}$$

Therefore $\mu(S_\delta) = 1$. Now, note that if $x \in S_\delta$, then there is $n_k^+ \to \infty$ for which for every k, $D_{n_k}^\xi(x) \in U_\delta$. This in turn implies that

$$\#\left\{0 \le i < n_k^+ \,\middle|\, S_i^\xi(x) > \sqrt{n_k^+}\right\} \ge (1-\delta)n_k^+.$$

Analogously, there is $n_k^- \to \infty$ such that for every k,

$$\#\left\{0 \le i < n_k^- \,\middle|\, S_i^\xi(x) < -\sqrt{n_k^-}\right\} \ge (1-\delta)n_k^-.$$

Both can be used as in the previous proof.

Corollary: north-south maps

Consider some diffeomorphisms $f_1, f_2, \ldots, f_n \in \mathrm{Diff}_{C^2}(I)$ preserving the orientation of the interval that commute with each other.

such that $f_+(y) > y$ and $f_-(y) < y$ for all $y \in (0,1)$.

We first notice that we can always find

The covering map p is isomorphic to the usual covering map $\pi\colon \mathbb{R} \to \mathbb{R}/\mathbb{Z} = \mathbb{S}^1$ via some homeomorphism $h\colon \mathbb{R} \to \mathring{I}$ such that $\pi = p \circ h$ [for instance, 11, Proposition 1.37].

Chapter 4

Partially hyperbolic dynamics

4.1 Hyperbolicity

Lemma 19 (λ-lemma)**.** *If p is a hyperbolic fixed point and D^u is a u-dimensional disk such that $D^u \pitchfork W^s(q)$, then for all unstable ball $B^u_R(p)$ and $\varepsilon > 0$,*

4.2 Revisiting

4.3 Partial hyperbolicity

4.4 Invariance Principle

Definition 6. A partially hyperbolic diffeomorphism is *accessible* if between any two points in the manifold, there is a path composed of segments tangent to either E^s or E^u.

Definition 7. A partially hyperbolic diffeomorphism is *dynamically coherent* if there exists invariant foliations $\mathcal{F}^{cs}$ and $\mathcal{F}^{cu}$ tangent to $E^c \oplus E^s$ and $E^c \oplus E^u$, respectively.

Theorem 20. *Let $f\colon M \to M$, where $M = X \times Y$, be a skew product diffeomorphism with an ergodic invariant measure μ. Then, $\lambda^c(\mu) \leq 0$ implies that $\{\mu^c_x\}_{x\in M}$ is invariant by unstable holonomies, meaning that $(h^u_{x,y})_*\mu^c_x = \mu^c_y$.*

We begin the proof of theorem 20 by examining a clever trick for establishing the invariance of the conditional measures by the unstable holonomy. Let

Appendix A

Measure and probability

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2. E. S. Andersen, "On the introduction of measures in infinite product sets," *Matematisk-Fysiske Meddelelser*, vol. XXV, no. 4,

A.1 Probability spaces

The Kolmogorov Extension Theorem

Products and limits

We can construct a product measure

A.2 Random variables

Independence and distribution

A.3 Weak convergence

Theorem 21 (Portmanteau)**.** *The following are equivalent:*

i. $\mu_n \xrightarrow{w} \mu$*;*

ii. $\lim_{n\to\infty} \mu_n(\phi) = \mu(\phi)$ *for all Lipschitz functions* ϕ*;*

iii. $\liminf_{n\to\infty} \mu_n(U) \geq \mu(U)$ *for all open sets* U*;*

iv. $\limsup_{n\to\infty} \mu_n(F) \leq \mu(F)$ *for all closed sets* F*;*

v. $\lim_{n\to\infty} \mu_n(A) = \mu(A)$ *for all continuity sets* A *of the measure* μ*.*

A.4 Ergodic theorems

Hewitt-Savage zero-one law

A.5 Brownian motion

The space of càdlàg functions

The space $C([0,1])$ of continuous functions does not let us talk about processes that contain discontinuities, or even how discontinuous processes can converge in a suitable way to continuous ones. This is important in our setting to deal with the convergence of certain processes induced by discrete dynamics to the Brownian motion, which is continuous almost surely.

For this, we shall introduce the space of *càdlàg* functions, or Skorohod space. This is the space of functions whose discontinuities are of the *first kind*, that is, both lateral limits exist, and whose value at the point is equal to the right limit:

$$\mathbb{D}([0,1]) = \left\{ f\colon [0,1] \to \mathbb{R} \,\middle|\, \forall t, \lim_{s\to t^-} x(s) \text{ exists and } \lim_{s\to t^+} x(s) = x(t) \right\}.$$

We want to put a metric on this space with the property that functions are close together if their values are close to each other uniformly, but also allowing small perturbations in the parametrization of time.

Definition 8 (Skorohod topology)**.** Let Λ be the set of strictly increasing continuous functions from $[0,1]$ to itself. For $f, g \in \mathbb{D}([0,1])$, we define

$$d(f,g) = \inf_{\lambda\in\Lambda} \max\left\{ \|\lambda - \mathrm{id}\|_\infty, \|f - g\circ\lambda\|_\infty \right\}.$$

The distance above defines a metric on $\mathbb{D}([0,1])$, whose topology is called the Skorohod topology. Since $d(f,g) \le \|f-g\|_\infty$, the Skorohod topology is coarser than the uniform topology.

Proposition 22. *The restriction of the Skorohod topology to $C([0,1])$ coincides with the uniform topology.*

Proof. Since the Skorohod topology is coarser, it remains to show that if $f, f_n \in C([0,1])$ and $d(f_n, f) \to 0$, then $\|f_n - f\|_\infty \to 0$. But note

$$\|f_n - f\|_\infty \le \inf_{\lambda\in\Lambda} [\|f_n - f\circ\lambda\|_\infty + \|f\circ\lambda - f\|_\infty] \to 0$$

by the convergence assumption and uniform continuity of f. □

The Wiener measure and arcsine law

insert properties (and construction?) of the Wiener measure $W \in \mathcal{M}_1(\mathbb{D}([0,1]))$.

Let $\tau^+\colon \mathbb{D}([0,1]) \to \mathbb{R}$ be the observable that returns the proportion of time its argument is positive, that is, $\tau^+(f) = \mathrm{Leb}\{x \in [0,1] : f(x) > 0\}$.

Lemma 23. *τ^+ is measurable and continuous in a full W-measure set.*

Proof. adapt this to $\mathbb{D}([0,1])$ This is proved in [10, appendix M15]. Since the normal distributions are non-atomic and for every $t \in [0,1]$, W_t has normal distribution, $W(\{f \in \mathbb{D}([0,1]) : f(t) = 0\}) = 0$ for all t, so by Fubini, $\mathrm{Leb}(\{t \in [0,1] : f(t) = 0\}) = 0$ for W-almost every f. It remains to check that τ^+ is continuous at such points $f \in W$. Indeed, 0 is the only discontinuity of the indicator function $1_{(0,\infty)}$, so if f is in this set and $f_n \to f$ uniformly, then $1_{(0,\infty)} \circ f_n \to 1_{(0,\infty)} \circ f$ almost everywhere, so by the DCT the same is true of their respective integrals. □

Theorem 24 (Arcsine law)**.** *For $\alpha \in [0,1]$,*

$$(\tau^+)_* W([0,\alpha]) = \frac{2}{\pi} \arcsin \sqrt{\alpha}.$$

Theorem 25 (Forgery theorem)**.** *If $f \in C([0,1])$ and $f(0) = 0$, then every neighborhood of f has positive Wiener measure.*

Weak invariance principles

We will fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ for the rest of this section. Given a sequence $\xi = (\xi_1, \xi_2, \dots)$ of random variables defined on Ω, we can define their partial sums $S_n^\xi = \sum_{i=1}^n \xi_i$ and a sequence of processes $D_n^\xi \colon \Omega \to \mathbb{D}([0,1])$ given by

$$D_n^\xi(\omega)(t) = \frac{S^\xi_{\lfloor nt \rfloor}(\omega)}{\sqrt{n}},$$

obtained by appropriately scaling the partial sums of the random variables up to some iterate n.

Remarkably, for many classes of sequences ξ, their corresponding processes D_n^ξ converge in some way to the Brownian motion — we denote by an *invariance principle* a result of this form. Perhaps the best known invariance principle is the Donsker's theorem, which is a *weak* invariance principle since it asserts convergence in the weak topology.

Theorem 26 (Donsker)**.** *If $\xi \subseteq L^2(\mathbb{P})$ is a sequence of i.i.d. variables with mean zero and variance 1, then the measures $(D_n^\xi)_* \mathbb{P}$ converge weakly to the Wiener measure. In other words, for every measurable continuous function $\varphi \colon \mathbb{D}([0,1]) \to \mathbb{R}$,*

$$\int \varphi \circ D_n^\xi \, d\mathbb{P} \to \int \varphi \, dW.$$

The motivation for looking into invariance principles is that they allow transferring certain results from Brownian motions to the sequence of partial sums S_n^ξ. In our case, the result we are interested in is the arcsine law. We know from lemma 23 that τ^+ is continuous a.e. , so weak convergence is enough to transfer its conclusions:

Lemma 27. *Suppose $\xi \subseteq L^2(\mathbb{P})$ is such that $(D_n^\xi)_* \mathbb{P} \xrightarrow{w} W$. Then the average of the time in which the random walk S_n^ξ is above $M \in \mathbb{R}$ converges to an arcsine distribution, in the sense that*

$$\lim_{n\to\infty} \mathbb{P}\left(\left\{\omega \in \Omega \;\middle|\; \frac{1}{n}\#\{1 \le i \le n \mid S_n^\xi(\omega) > M\} \le \alpha\right\}\right) = \frac{2}{\pi} \arcsin \sqrt{\alpha}.$$

Proof. Writing $\tau_n^+(\omega) = \frac{1}{n}\#\{1 \le i \le n \mid S_n^\xi(\omega) > M\}$, the left-hand side can be rewritten as $\lim_{n\to\infty}(\tau_n^+)_*\mathbb{P}([0,\alpha])$. Set $c_n = M/\sqrt{n}$, and note that

$$\tau_n^+(\omega) = \frac{1}{n}\#\left\{1 \le i \le n \;\middle|\; \frac{S_i^\xi(\omega)}{\sqrt{n}} - \frac{M}{\sqrt{n}} > 0\right\} = \tau^+(D_n^\xi(\omega) - c_n).$$

Since by theorem 24 we know $Z = (\tau^+)_*W$ is equivalent to the Lebesgue measure on $[0,1]$, intervals are continuity sets for it. By the weak convergence assumption and continuity of τ^+, we know $(\tau^+ \circ D_n^\xi)_*\mathbb{P} \xrightarrow{w} Z$, and since $c_n \to 0$ (as a sequence in $\mathbb{D}([0,1])$), in fact $(\tau^+ \circ (D_n^\xi - c_n))_*\mathbb{P} \xrightarrow{w} Z$. By Portmanteau, this means

$$\lim_{n\to\infty}(\tau_n^+)_*\mathbb{P}([0,\alpha]) = \lim_{n\to\infty}(\tau^+ \circ (D_n^\xi - c_n))_*\mathbb{P}([0,\alpha]) = \frac{2}{\pi}\arcsin\sqrt{\alpha}.$$

□

Lemma 28. *Under the same hypothesis as lemma 27, for all $M \in \mathbb{R}$ and for $\mathbb{P}$-almost every $\omega \in \Omega$,*

$$\lim_{n\to\infty}\frac{1}{n}\#\{1 \le i \le n \mid S_i^\xi(\omega) \in [-M,M]\} = 0.$$

Proof. Just like in the previous proof, write $\tau_n(\omega) = \frac{1}{n}\#\{1 \le i \le n : \mid S_n^\xi(\omega) \in [-M,M]\}$, and then note

$$\tau_n(\omega) = \tau^+(D_n^\xi(\omega) + c_n) - \tau^+(D_n^\xi(\omega) - c_n).$$

where $c_n = M/\sqrt{n}$. So

$$\lim \int \tau_n \, d\mathbb{P} = \int \tau^+ \, dW - \int \tau^+ \, dW = 0.$$

Finish!! □

Ergodic Martingale Differences

Donsker's theorem is far from being the most general result of its kind, and in our case, we are interested in a generalization of such weak invariance principle for ergodic martingale difference sequences. Intuitively, these are sequences that behave as the time-one steps of a martingale, in the sense that even with information about the previous steps the expected value of the next step is still zero.

Definition 9. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables is said to be a *martingale difference sequence* if for all n,

$$\mathbb{E}_\mathbb{P}(\xi_n \mid \sigma(\xi_1, \dots, \xi_{n-1})) = 0.$$

Stationarity and two-sided processes

In the previous definition we considered an one-sided sequence of random variables. As it is often the case in ergodic theory, the sequences we shall be dealing with will have the additional property of *stationarity*, which amounts to some sort of time-invariance. This property will also allow us to lift one-sided sequences of variables to two-sided ones, that is, sequences $(\dots, \xi_{-1}, \xi_0, \xi_1, \dots)$ with indices in $\mathbb{Z}$, due to the existence of invertible extensions as exposed in section 1.2 (which ultimately boils down to the Kolmogorov Extension Theorem).

Definition 10. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables taking values in Y is said to be *stationary* if $\xi_*\mathbb{P}$ is an invariant measure for the left shift $\sigma\colon Y^{\mathbb{N}} \to Y^{\mathbb{N}}$. Similarly, we can define stationarity for two-sided sequences by replacing $\mathbb{N}$ by $\mathbb{Z}$ in the definition.

Lemma 29. *Every stationary one-sided ergodic martingale difference sequence ξ is associated to a two-sided stationary ergodic martingale difference sequence ζ with the same marginal distributions on positive indices. That is, if $\pi\colon Y^{\mathbb{Z}} \to Y^{\mathbb{N}}$ is defined by $\pi(\dots x_{-1}, x_0, x_1, \dots) = (x_1, x_2, \dots)$, we have $(\pi \circ \zeta)_*\mathbb{P} = \xi_*\mathbb{P}$.*

Remark 7. Test

Invariance principle for ergodic sequences

Definition 11. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables is said to be *ergodic* if $\xi_*\mathbb{P}$ is an ergodic measure for the left shift $\sigma\colon Y^{\mathbb{N}} \to Y^{\mathbb{N}}$.

The next result can be found in [10, theorem 18.3].

Theorem 30. *If $\xi \subseteq L^2(\mathbb{P})$ is a stationary ergodic martingale difference sequence for which $\sigma = \|\xi_1\|_2 > 0$, then the measures $(D_n^{\xi})_*\mathbb{P} \xrightarrow{w} \sigma W$.*

Reverse martingale difference sequences

In ergodic dynamics, the processes we usually care about are the values $\xi = (\varphi \circ f^{n-1})_{n\geq 1}$ of some observable $\varphi \in L^1(\mu)$ along orbits of f. This process, however, cannot possibly be a martingale difference sequence unless $\varphi \equiv 0$, because $\varphi \circ f^n$ is measurable for the σ-algebra $f^{-n}(\mathcal{F})$, which is non-increasing due to measurability of f. In other words, we always have $\mathbb{E}_\mu(\varphi \circ f^n \mid \sigma(\varphi, \dots, \varphi \circ f^{n-1})) = \varphi \circ f^n \neq 0$ in general. This motivates the following definition.

Definition 12. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables is said to be a *reverse martingale difference sequence* if for all n,

$$\mathbb{E}_{\mathbb{P}}(\xi_n \mid \sigma(\xi_{n+1}, \xi_{n+2}, \dots)) = 0.$$

In the case of a stationary sequence (as is the case for $\xi = (\varphi \circ f^{n-1})_{n\geq 1}$ and $\mathbb{P} = \mu$ an invariant measure for f), results for martingale difference sequences can be stated dually for reverse martingale difference sequences. The following discussion expands on comments and text present in [10, Appendix M22 and Section 19].

Corollary 31. *If $\xi \subseteq L^2(\mathbb{P})$ is a stationary ergodic reverse martingale difference sequence for $\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \cdots$ for which $\sigma = \|\xi_1\|_2 > 0$, then the measures $(D_n^{\xi})_*\mathbb{P} \xrightarrow{w} \sigma W$.*

Proof. Consider the space $\mathbb{R}^{\mathbb{N}}$ with the product σ-algebra $\mathcal{H} = \mathcal{B}^{\otimes\mathbb{N}}$ and the measure $\nu = \xi_*\mathbb{P}$. This measure is invariant by the left shift $\sigma\colon \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}}$ by the definition of stationarity. Since $\mathcal{H}$ coincides with the Borel σ-algebra, the construction of the invertible extension (see section 1.2) gives us a stationary measure $\tilde{\nu} \in \mathcal{M}(\mathbb{R}^{\mathbb{Z}})$ for the invertible shift, which we shall call $\tilde{\sigma}\colon \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{Z}}$,

whose projection $\pi_* \tilde{\nu}$ by $\pi \colon \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{N}}$ is equal to ν. The measure $\tilde{\nu}$ is also ergodic for the invertible shift, since ν is.

We now construct an (usual) martingale difference sequence on this space. Let $\mathcal{F}_n = \pi^{-1}(\mathcal{G}_n)$

Finish!! $\square$

Appendix B

Some categorical language

Objects

W 31

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